

Contacts:

SCCAF Process : Graham Monks : gmonks

STA : Sunita Shahmehri : sshahmeh

Internal Manufacturing:

IMG: Amr Hegazy : ahegazy

NA: Leah May : lmay

Europe: Alexander Clemens : acleme36

China: Rui Ma : rma7

PT Internal SCCAF : Sandy Kenney : slee27

SCCAF Handbook

User Guide to the SCCAF Process

(**S**pecial **C**haracteristics **C**ommunication & **A**greement **F**orm)

[Ford Automotive Procedure FAP-03-111](#)

Selection, Identification, and Control of Special Characteristics

Ford Access: [Global FMA Website - SCCAF](#)

[Global FMA Website](#)

[SCCAF External Reference Guide](#)

[SCCAF Internal Reference Guide](#)

[Life of a Special Characteristic - External](#)

[APQP PPAP Evidence Workbook Statistical Reference Guide](#)

Supplier Access: [Global FMA Website - SCCAF](#)

[Global FMA Website](#)

[SCCAF External Reference Guide](#)

[Life of a Special Characteristic - External](#)

SCCAF Handbook can be shared with Suppliers

For Supplier access see: https://web.qpr.ford.com/sta/Phased_PPAP.html

[SCCAF Handbook Change Log](#)

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SCCAF (Special Characteristics Communication and Agreement Form) is the form used for the process outlined in **Ford Automotive Procedure FAP-03-111 - Selection, Identification, and Control of Special Characteristics**.

The SCCAF is used to communicate initial Special Characteristics and associated control methods prior to the release of the approved Engineering drawing / CAD data and specifications for the purpose of manufacturing process planning and Supplier quotation.

In order to achieve the highest quality, lowest total cost and align with the timing identified in GPDS, Product Development must work concurrently with Manufacturing and Assembly activities to design, develop and optimize both product design and associated manufacturing processes.

The key outputs of this collaborative process are an optimized process design, an optimized product design, and a cross-functionally agreed initial list of Special Characteristics that require ongoing measurement and special controls during mass production.

In Summary, a SCCAF:

- Describes management of Special Characteristics.
- Includes Product Characteristics identified in the DFMEA.
- Documents control methods identified later in Process Control Plans & in the PFMEA.
- Is completed by GPDS Vehicle & Powertrain Unit deliverables:
 - **UNV1** : VG-T-72-D3 & **UPV1** : VG-T-24-D3
 - **UNV2** : VG-T-56-D3 & **UPV2** : VG-T-04-D3
 - **Unit PTC** : PO-T-40-D3
 - **Unit TPVDJ** : PO-T-40-D10

In Summary, an External SCCAF:

- Is an initial agreement between PD, STA and Supplier for the identification and manufacturing control of Special Characteristics (Superseded by Released CAD Data & Production Process Control Plan).
- Ford LEA D&R Engineer is responsible for completing the LHS (yellow sections) of the form, the overall SCCAF process and approval (see [p5 – Roles & Responsibilities](#)).
- Supplier completes RHS (green sections) of the form and approves complete SCCAF content.

In Summary, an Internal SCCAF:

- Is an initial agreement between PD and Ford Manufacturing for the identification and manufacturing control of Special Characteristics (Superseded by Released CAD Data & Production Process Control Plan).
- Details the Product Characteristics that require Special Controls.
 - Ford Final Vehicle Assembly Plant.
 - Ford Internal Manufacturing such as Stamping, Transmission or Engine Plant.
- Ford LEA D&R Engineer is responsible for completing the LHS (yellow sections) of the form, the overall SCCAF process and approval (see [p5 – Roles & Responsibilities](#)).
- Ford Manufacturing completes RHS (green section) of the form & approves complete SCCAF content.

Why do we need it?

- By identifying and agreeing to the Special Characteristics, we potentially reduce the risk of recalls, warranty and loss of customer satisfaction by having controls in place for all the important Design Characteristics listed.
- Drives alignment among all stakeholders in the process and also ensures that all manufacturing facilities have sufficient and agreed Process Control Methods ready for production.
- Is critical for all manufacturing locations, especially in emerging markets with a developing Supplier base & new Ford Manufacturing Plants to ensure that agreement on best practices for process methods are applied.
- There is a critical need to use a SCCAF to communicate efficiently and effectively.

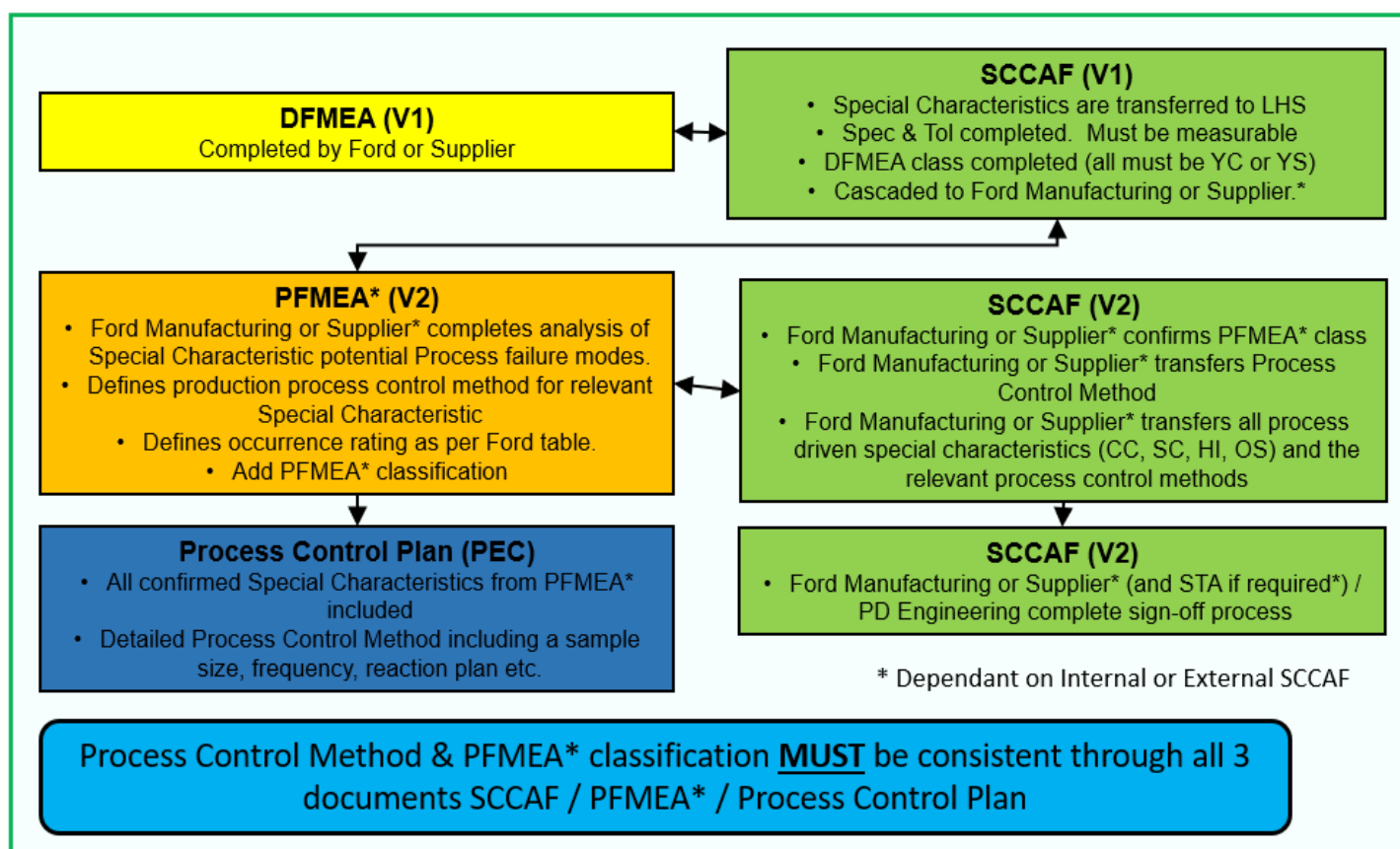
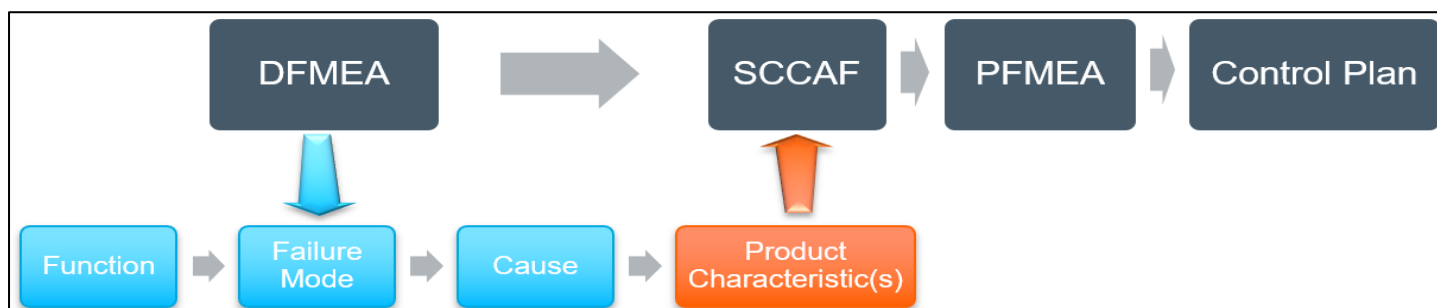
IS

- The vital few Special Characteristics.
- An initial agreement (handshake) between Design and Manufacturing on Characteristics and Controls.
- The basic details of Process controls (full details are in the control plan).
- The ongoing controls that are part of the regular process.
- A Point in Time for defined life.
- Superseded by the Production Control Plan.
- Initiated as a direct output of the DFMEA

IS NOT

- Everything we need the Supp to do.
- A list of manufacturing process steps.
- A method for cascading design or testing requirements to Manufacturing.
- A Substitute for ESOW.
- A Substitute for Engineering Specification (ES).
- An exhaustive list of Design Requirements.
- A list of DV & PV requirements.
- A living Document.
- A PFMEA.
- A list of defects to avoid.
- A BOM.
- Stand alone document.
- The only signed & agreed document on process.

Linkage to PFMEA & Process Control Plan



LEA D&R Engineers:

- Own the overall SCCAF process for all Manufacturing locations for both Internal and External SCCAFs.
- Cannot delegate this responsibility to regional engineers or joint venture engineers, as they do not have the required design knowledge. May delegate site visits to a local representative, but the LEA D&R Engineer must review the outcome & documents with the local representative and have final approval responsibility.
- Is responsible for completion to GPDS gateways for Internal and External SCCAFs, even if the Supplier is design responsible for the DFMEA. Must work in conjunction with Ford Manufacturing and Suppliers to complete SCCAFs. Is responsible to review and approve Supplier developed DFMEAs.
- Approve completed SCCAF in conjunction with STA, Supplier and Craftsmanship Engineer if required.
- Participate in the overall APQP (Schedule A) process.
- The approval and sign-off of the PFMEA and Process Control Plans for inverted delta parts.

STA Site Engineer:

- Assisting selected Suppliers in the overall APQP (Schedule A) and PPAP process.
- Assisting selected Suppliers in the process related elements as required (e.g. PFMEA, Control Plan).
- In conjunction with the LEA D&R Engineer, evaluate the proposed process control methods and approve the SCCAF for PPAP Submission Level 3, Level 5, and Priority Suppliers.

If requested by LEA D&R Engineer will also do the same regardless of the PPAP Submission Level.

Note: PPAP Submission Levels 2 and 4 are not used within Ford.

For PPAP Level definitions review the PPAP AIAG standard.

- The approval and sign-off of the PFMEA and Process Control Plans for inverted delta parts.

Supplier Design Lead (owns DFMEA):

- Works with Ford LEA D&R engineer to identify potential special characteristics from the DFMEA and transfer to the SCCAF.
- Identifying controls which are appropriate to meet the requirements of an SC or CC Characteristic and complete the Process Control Methods column.
- Add any process related Characteristics that are required to ensure robust delivery, including appropriate associated control methods.
- Ensure that the PFMEA / Process Control Plans align to the DFMEA severity ratings and effects.
- Ensure any product design improvements that could make the manufacturing process more robust are identified and communicated to Ford during regular team communication channels for SCCAF review.
- Approve completed SCCAF in conjunction with LEA D&R Engineer, STA and Craftsmanship Engineer if required.

Supplier Manufacturing (make to print):

- Identifying controls which are appropriate to meet the requirements of an SC or CC Characteristic and complete the Process Control Methods column.
- Add any process related Characteristics that are required to ensure robust delivery, including appropriate associated control methods.
- Approve completed SCCAF in conjunction with LEA D&R Engineer, STA and Craftsmanship Engineer if required.

Ford Internal Manufacturing:

- Identifying controls which are appropriate to meet the requirements of an SC or CC Characteristic and complete the Process Control Methods column.
- Add any process related Characteristics that are required to ensure robust delivery, including appropriate associated control methods.
- Ensure that the PFMEA / Process Control Plans align to the DFMEA severity ratings and effects.
- Ensure any product design improvements that could make the manufacturing process more robust are identified and communicated.
- Approve the completed SCCAF in conjunction with LEA D&R Engineer, Ford Plant Quality Manager and Craftsmanship Engineer if required.

GPDS deliverables (Vehicle & PT Unit)

VG-T-72-D3 - Under Body V1 Failure Mode Avoidance Status

VG-T-56-D3 - Under Body V2 Failure Mode Avoidance Status

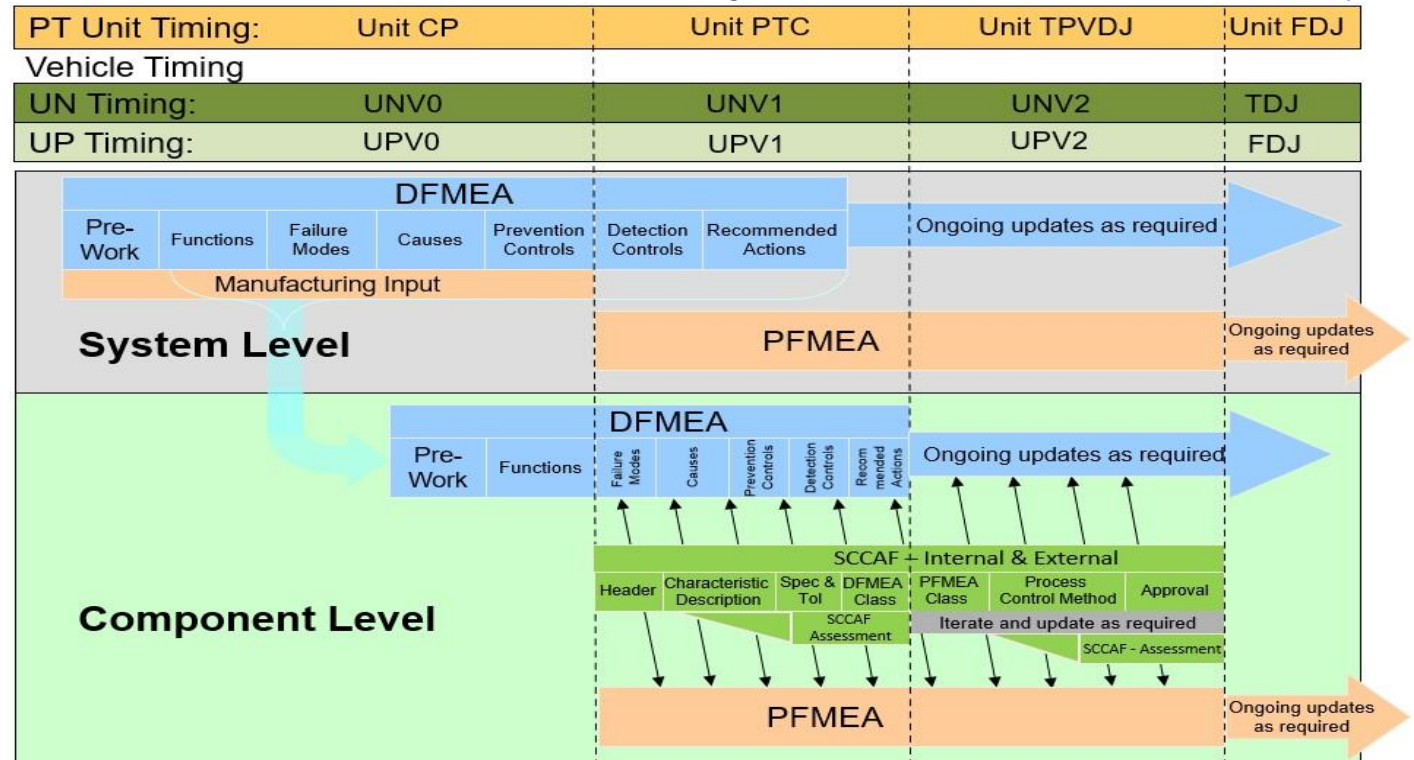
VG-T-24-D3 - Upper Body V1 Failure Mode Avoidance Status

VG-T-04-D3 - Upper Body V2 Failure Mode Avoidance Status

PO-T-40-D3 - Unit PTC Failure Mode Avoidance Status

PO-T-40-D10 - Unit TPVDJ Failure Mode Avoidance Status

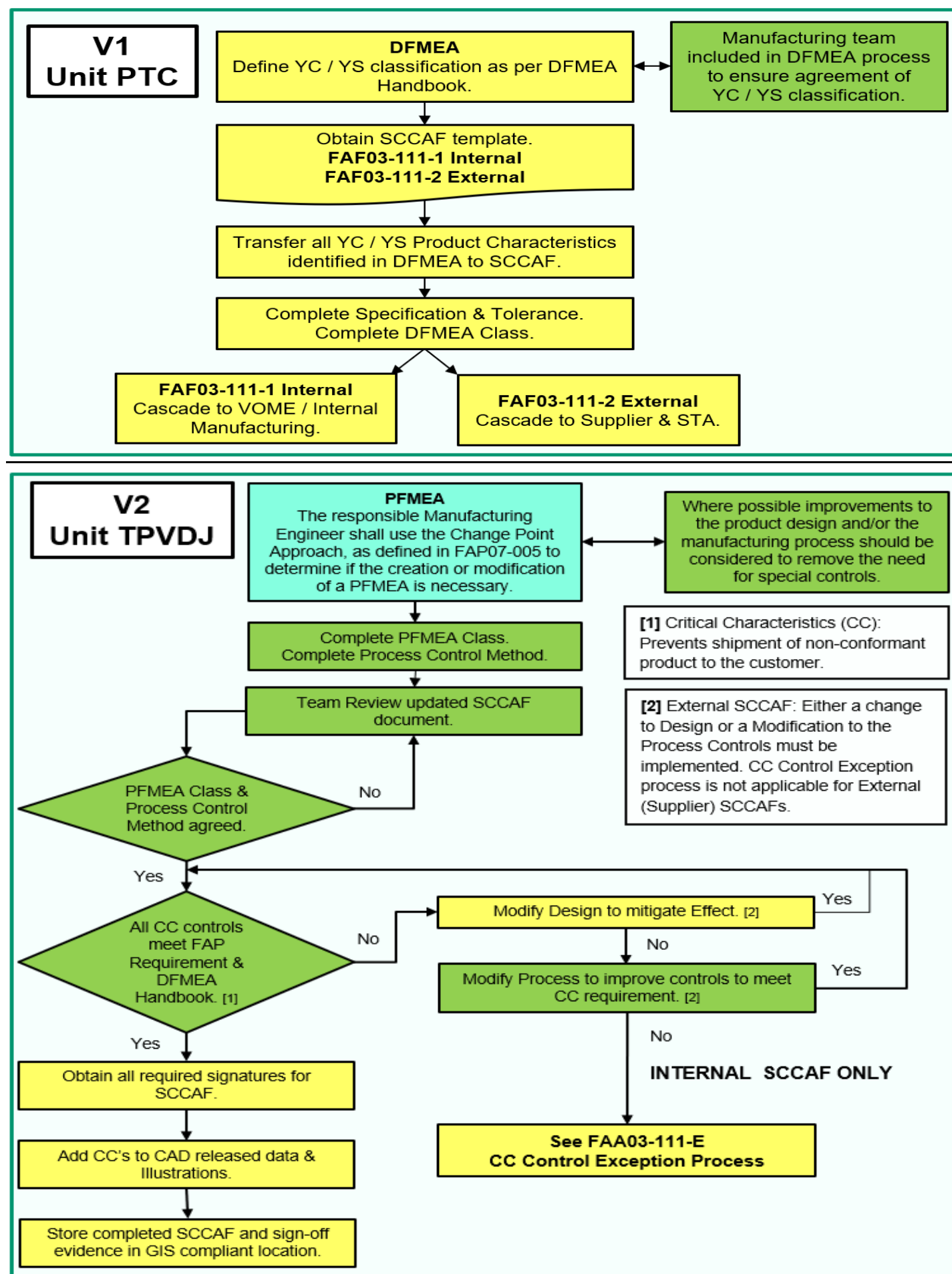
For more information on all FMA GPDS Deliverables including SCCAF see [FMA Deliverables Table - GPDS Gateways](#).



The DFMEA determines which Product Characteristics should be controlled based on the product design and consideration of the manufacturing processes used to produce it. The key process steps outlined below require a two-way communication between PD and Manufacturing:

- The DFMEA & PFMEA should be executed and updated as part of simultaneous engineering activity.
- The PFMEA analyses all steps of the manufacturing process to determine where Special Process Controls are needed in order to produce products that meet the design intent.
- Where possible, improvements to the product design and/or the manufacturing process should be considered to remove the need for Special Controls.
- Collaborative process shall include STA for External (Supplier) SCCAFs, including the Special Characteristics cascade at GPDS Gateway V1 / Unit PTC, to ensure that the Process Control Methods are adequate to meet the CC/SC criteria and process robustness.
- The SCCAF is completed and fully approved by the end of GPDS Engineering Gateway V2 / Unit TPVDJ based on the most current engineering concept design. This allows the SCCAF to be completed and available in time for consideration when determining initial program cost and manufacturing assumptions both internally and with external Suppliers.
- SCCAF Assessment is for External SCCAFs only and not required for Internal SCCAFs. Best practice is to complete the SCCAF and SCCAF assessment (if External SCCAF) progressively through the V2 / Unit TPVDJ milestone; this will provide the appropriate amount of time to understand the manufacturing process and the key process input variables which govern the outcome of characteristics rather than relying on monitoring characteristics with EOL checks.
- Once the SCCAF has been fully approved at V2 / Unit TPVDJ the SCCAF will be imported into the "APQP/PPAP Evidence Workbook" which is used to document all measurement data, gauging, capability information related to each characteristic through the launch process.

LEA D&R Engineer has overall responsibility for SCCAF Process (see [p5](#) for further details)



The FAP03-0111 Selection, Identification, and Control of Special Characteristics does not include an update process.

After the SCCAF is signed at V2 / Unit TPVDJ gateway, a SCCAF would only require an update & re-sign if:

- The Engineering team identifies new or obsolete characteristics through DFMEA updates.
- Process modifications require changes to agreed controls prior to PPAP, update is not required if these controls are additional to already agreed Process Control methods (D&R Engineer should be approving all changes to the latest Control plan).

Any changes to existing information contained in the SCCAF post “xDJ” are controlled through the Drawing (WERS release) and Process Control Plan via a SREA after black-out period.

Where the only modification to the SCCAF is the “Specification and Tolerance” an update is not required as per the Ford FAP process, but teams may update if they wish. The documents that must align are the Released CAD Data / Drawing and the Supplier PFMEA and Process Control Plans.

Where Spec & Tolerance differ in SCCAF to latest Released CAD data / Drawing, the released data is Master.

Below is a table showing possible Design or Process Changes and the required documents to be updated after approval @ V2 / Unit TPVDJ.

Change Type	Program Stage	Required Ford Process to Control Change	Documents / Data to be modified and re-approved				
			DFMEA	SCCAF	CAD Model / Drawing	PFMEA	Process Control Plan
New Special Characteristic identified in DFMEA - YC / YS	Pre P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes
	Post P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes
Design Change to remove Special Characteristic in DFMEA - YC / YS	Pre P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes
	Post P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes
Modification to Specification or Tolerance of a Product Characteristic	Pre P-Release	Evidence Work Book	No *	No	Yes	Yes	Yes
	Post P-Release	WERS Concern	No *	No	Yes	Yes	Yes
Supplier request for Modification of Process Control Method to that agreed in SCCAF	Pre PPAP	SCCAF	No	Yes	No	Yes	Yes
	Post PPAP	SREA	No	No	No	Yes	Yes
Supplier Request for removal of SC due to demonstrated process capability with data	Pre Job 1	Not Permitted - Production data required to prove Process Capability					

* Team should review the DFMEA using the Change Point Approach to understand if the change requires an update as per FAP07-005.

Engineering Change post Job 1

For a design change to an existing component or assembly, the Engineering team introducing the change should review the DFMEA and relevant FMA Documentation using the Change Point Approach.

- If the design change identifies new or redundant Potential Special Characteristics (YC / YS) the SCCAF must be updated and resigned by all parties, as per the table above.
- Where SCCAF update is required all SCCAF approvals should be obtained prior to the final CAD Data / Drawing release in WERS.
- The engineering release number related to the engineering change is entered into the “Engineering Release Number” field in the SCCAF Header.

Change Type	Change	Required Ford Process to Control Change	Documents / Data to be modified and re-approved				
			DFMEA	SCCAF	CAD Model / Drawing	PFMEA	Process Control Plan
Design Change - Post Job 1	Change to Potential Special Characteristics - YC / YS	WERS Concern	Yes	Yes	Yes for CC only	Yes	Yes
	Change to Specification or Tolerance of a Product Characteristic	WERS Concern	No *	No	Yes	Yes	Yes
New or Changed Process - Post Job 1	Request for Modification of Process Control Method to that agreed in SCCAF	SREA	No	No	No	Yes	Yes
	Request for removal of SC due to demonstrated process capability with data	SREA	No	No	No	Yes	Yes

* Team should review the DFMEA using the Change Point Approach to understand if the change requires an update as per FAP07-005.

The FAP03-0111 Selection, Identification, and Control of Special Characteristics does not include an update process.

After the SCCAF is signed at V2 / Unit TPVDJ gateway, a SCCAF would only require an update & re-sign if:

- The Engineering team identifies new or obsolete characteristics through DFMEA updates.
- Process modifications require changes to agreed controls prior to P-Release, update is not required if these controls are additional to already agreed Process Control methods (Product Engineering should be approving all changes to the latest Control plan).

Any changes to existing information contained in the SCCAF post “xDJ” are controlled through the Drawing (WERS release) and Process Control Plan.

Where the only modification to the SCCAF is the “Specification and Tolerance” an update is not required as per the Ford FAP process, but teams may update if they wish. The documents that must align are the Released CAD Data / Drawing, the PFMEA (if applicable) and Process Control Plans.

Where Specification & Tolerance differ in SCCAF to latest Released CAD data / Drawing, the released data is Master.

Below is a table showing possible Design or Process Changes and the required documents to be updated after approval @ V2 / Unit TPVDJ.

		Required Ford Process to Control Change	Documents / Data to be modified and re-approved					
Change Type	Program Stage		DFMEA	SCCAF	Drawing / CAD Model & PD Illustrations	GSPAS (VO) (If Applicable)	PFMEA (If Applicable)	Control Plan
New Special Characteristic identified in DFMEA - YC / YS	Pre P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes	Yes
	Post P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes	Yes
Design Change to remove Special Characteristic in DFMEA - YC / YS	Pre P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes	Yes
	Post P-Release	SCCAF	Yes	Yes	Yes for CC only	Yes	Yes	Yes
Modification to Specification or Tolerance of a Product Characteristic	Pre P-Release	SCCAF	No *	Yes	Yes	Yes	Yes	Yes
	Post P-Release	WERS Concern	No *	No	Yes	Yes	Yes	Yes
Request for Modification of Process Control Method to that agreed in SCCAF	Pre P-Release	SCCAF	No	Yes	No	Yes	Yes	Yes
	Post P-Release	^Control Plan (VO) or SCCAF (PT)	No	No (VO) Yes (PT)	Yes (If Applicable)	Yes	Yes	Yes
Request for removal of SC due to demonstrated process capability with data	Pre Job 1	Not Permitted - Production data required to prove Process Capability						
^ Signed off control plan								
* Team should review the DFMEA using the Change Point Approach to understand if the change requires an update as per FAP07-005.								

Engineering Change post Job 1

No Change to the SCCAF is required post Job 1

Change Type	Change	Required Ford Process to Control Change	Documents / Data to be modified and re-approved					
			DFMEA	SCCAF	Drawing / CAD Model & PD Illustrations	GSPAS (VO)	PFMEA (If Applicable)	Process Control Plan
Design Change - Post Job 1	Change to Potential Special Characteristics - YC / YS	WERS Concern	Yes	No	Yes for CC only	Yes	Yes	Yes
	Change to Specification or Tolerance of a Product Characteristic	WERS Concern	No*	No	Yes	Yes	Yes	Yes
New or Changed Process - Post Job 1	Request for Modification of Process Control Method to that agreed in SCCAF	WERS Concern	No	No	No	Yes	Yes	Yes
	Request for removal of SC due to demonstrated process capability with data	WERS Concern	No	No	No	Yes	Yes	Yes

* Team should review the DFMEA using the Change Point Approach to understand if the change requires an update as per FAP07-005.

A new SCCAF **is** required for:

- All NTEI's (New Tooled End Item) in a program, regardless of level of change.
- Each manufacturing location of component/assembly, even if it is manufactured by the same Supplier in different locations.
- Re-sourcing of a Carry-Over component even if it is maintaining a carry-over part number.
- Change of manufacturing location for an existing Supplier even if it is maintaining a carry-over part number.

Note: For Internal SCCAF : See more detailed information on when required ([p11](#)).

A new SCCAF **is not** required for:

- Carry-Over component from previous program and Carry-Over Supplier/ Manufacturing location as the Process Control Plan for that component will already exist and therefore no requirement for an External SCCAF to cascade Special Characteristics.

Do you require SCCAF?	Yes	No
Carryover component and Carryover supplier manufacturing location		✓
Carryover component and New supplier Manufacturing location	✓	
New Component (New Part Number) – YS / YC identified in DFMEA	✓	

The number of Supplier manufacturing locations determines the number of External SCCAFs.

A SCCAF is required for each manufacturing location regardless of whether the same Supplier is used. This is required to make sure the Process Control Method is detailed and agreed for each individual location, as these methods may vary even within the same Supplier.

The Process Control method and PFMEA must be created for each manufacturing location even if the Technical Lead that develops and reviews the SCCAFs is from a Central Engineering Centre.

Approval of completed SCCAF is completed by local site STA (when requested) and the Supplier Plant Quality Manager (mandatory) for each manufacturing location.

How Many External SCCAFs	1	2	Example
1 Supplier & single manufacturing location	✓		John Doe Inc. : Pleasantville Plant : USA
1 Supplier & 2 different manufacturing locations		✓	John Doe Inc. : Pleasantville Plant : USA John Doe Inc. : Chennai : India
2 different Suppliers (therefore 2 different manufacturing locations)		✓	John Doe Inc. : Pleasantville Plant : USA Smith Engineering : Cologne : Germany

Note: In the unlikely event that a part has No potential special characteristics and No confirmed Special Characteristics, the team should note under the Characteristic Description “No characteristics identified” and complete the approval process for the blank SCCAF. As per PPAP Customer Specific Requirements, where non-special characteristics are designated for process capability demonstration, those characteristics should be recorded on the SCCAF.

External SCCAFs are very clear in the guidance that if there is a new part number or change of manufacturing location then a new SCCAF is required; Internal SCCAFs are not always as simple and clear cut.

Issues often arise when the components are completely Carry-Over but are adopted from another program. For example, a Transit wiper motor with no change (carryover part number) that is now being used on Explorer or with minor changes such as software.

Below is some basic guidance on when you require an Internal SCCAF:

- If part is C/O and Supplier or Ford Manufacturing location is the same as the current production on existing program:
 - **No**, as Process Control Plan will already exist.
- If part is C/O Supplier & Supplier Manufacturing location but component is from another program:
 - **Yes** if:
 - New characteristic needs to be controlled by the Ford plant assembling new component.
 - Specification and Tolerance changes from an existing similar part to new component such as fixing torque.
 - Requires different Bill of Process.
 - New manufacturing line.
- All NTEIs with new part number (prefix-base-suffix) require a SCCAF for a new program even if the part is only a minor change. A new internal SCCAF is still required even if it will be a carryover process and values (such as torque) within Internal Manufacturing.
Teams should review with regional Internal Manufacturing Quality or regional FMA teams as some exceptions to this may be acceptable (e.g. colour change only).
- For a CPSC level SCCAF when only certain line items are changing (Brake lines SCCAF where only one brake line has changed on a new program) a new SCCAF is still required and all items need to be cascaded including carry over line items.
This ensures PD has cascaded everything required and helps with future programs to ensure no items are not tracked.

Note: Teams need to consider whether an individual SCCAF is required just for this component or whether this can be included in combined Internal SCCAF for the relevant CPSC.

Internal SCCAFs are often at the System level due to the small number of characteristics for each component so teams can combine line items into one Internal SCCAF for a group of related components within a CPSC. This analysis would normally be done at xVO / Unit CP, when first analysing the number of SCCAFs required for a program. Agreement can be made within the PD Engineering teams on how many Internal SCCAFs are required and how the team are going to handle the Internal SCCAFs within their functional teams.

What is important is that “Design Responsibility” or “SALT Class” is not changed to suit a situation as there is a risk of a potential GAO Audit failure.

The number of Internal SCCAFs is determined by the number of Ford manufacturing locations.

Internal SCCAFs can be for Ford Final Assembly plants such as Dearborn Truck plant or Valencia but also be for Ford Stamping, Engine or Transmission plants.

How Many Internal SCCAFs	2	3	Example
1 Supplier and single manufacturing location supplies to 2 Ford plants	✓		John Doe Inc. : USA to Michigan Assembly Plant : USA John Doe Inc. : USA to FTM Plant : Thailand
2 Suppliers and manufacturing locations that supply to 3 Ford plants		✓	John Doe Inc. : USA to Pacheco : Argentina Smith Engineering : Germany to Chongqing : China Smith Engineering : Germany to Valencia : Spain

The global templates are available from:

- [Global FMA Website - SCCAF](#)
- [FAP03-111 : Selection, Identification, and Control of Special Characteristics](#)

There are only 2 acceptable SCCAF forms:

- [FAF03-111-1-Internal Ford Manufactured and Assembled Content](#)
 - To be used for Ford Final Assembly plants (VOME) and other Ford Manufacturing locations including Stamping, Engine, and Transmission etc.
- [FAF03-111-2-Externally Supplied Content](#)
 - To be used for all Supplier commodities, including components, assemblies and Supplier assembled modules.

These forms must be used without substitution. The SCCAF template/form from the “APQP/PPAP Evidence Workbook”, Supplier variants or old Supplement K documents are not accepted. The form shall **not** be modified in any way, as this is a Global Template, except to add rows for additional Product Characteristics when required.

For External SCCAF : Once the SCCAF has been fully approved at V2 / Unit TPVDJ the SCCAF will be imported into the “APQP/PPAP Evidence Workbook” which is used to document all measurement data, gauging, capability information related to each characteristic through the launch process.

The correct specific form (Internal or External) must be used to ensure the relevant local plant location and signatories complete the approval.

FAF03-111-1 : Internal Ford Manufactured and Assembled Content

FAF03-111-1 v2.3 March 2023					Special Characteristics Communication and Agreement - Internal				OPTIONAL: SCCAF Draft Vers: Date:		
Lead / Support roles and timing for each stage		Ford PD engineer / Mfg		Manufacturing / Ford PD		SCCAF Handbook		SCCAF Internal Reference Guide		Special Characteristics for:	
Model Year & Program Code		2021 Scorpion 6.7		Part / System Number		BC3Q-6153-A		Approval Signature		Name	
Ford Internal Manufacturing Plant		Rawsonville - USA		Engineering Release Number				Ford D & R Engineer Approval / Date		AN Other	
Ford Plant GSDB Code				Design Lead		Ford		Ford Manufacturing Engineer Approval / Date		Ann Engineer	
Part / System Name		Ring - Oil Control		Ford Customer Plant		Saarlouis Assembly Plant - Germany		Ford Craftsmanship Approval / Date			
CPSC / COE / UN-UP		050505 VHM3 UN		DFMEA Reference		2021 B123 - Oil Control		Ford Plant Quality Manager Approval / Date		A Part	
Note: Approvals may be obtained by physical signature or electronic sign-off. Recommended: Use DocuSign to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment (See SCCAF FMA webpage for DocuSign help).											

FAF03-111-2 : Externally Supplied Content

FAF03-111-2 v2.3 March 2023					Special Characteristics Communication and Agreement - External				OPTIONAL: SCCAF Draft Vers: Date:		
Lead / Support roles and timing for each stage		Ford PD engineer / Supplier		Supplier / Ford PD		SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section		SCCAF Handbook - SCCAF Global FMA Website		Special Characteristics for:	
Model Year & Program Code		2021 B123 MCA		Part / System Number		BC3Q-6153-A		Approval Signature		Name	
Supplier Name / Plant		John Doe Inc. - Pleasantville Plant - USA		Engineering Release Number				Ford D & R Engineer Approval / Date		AN Other	
Supplier GSDB Code		A1234		Design Lead		Ford		Ford STA Engineer Approval / Date		Ann Engineer	
Part / System Name		Ring - Oil Control		Ford Customer Plant		Saarlouis Assembly Plant - Germany		Ford Craftsmanship Approval / Date			
CPSC		050505		DFMEA Reference		2021 B123 - Oil Control		Supplier Plant Quality Manager Approval / Date		A Part	
Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).											

Optional SCCAF Draft Version & Date Box

This can be used to help with version control of the draft SCCAF before final signoff. After the SCCAF is signed at the V2 / Unit TPVDJ gateway, all changes are controlled through the Drawing (WERS release) and Process Control Plan (SREA or WERS Concern). The SCCAF has no update control after the V2 / Unit TPVDJ sign off.

OPTIONAL: SCCAF Draft Vers: Date:		
Special Characteristics for:		
2021 B123 MCA	John Doe Inc. - Pleasantville Plant - USA	BC3Q-6153-A

FAF03-111-1 v2.3 March 2023

Special Characteristics Communication and Agreement - Internal

OPTIONAL: SCCAF Draft Vers: Date:

1. DFMEA & potential special characteristics

Type ONLY in Yellow Boxes

1A. PFMEA & Special Characteristics Agreement

Type ONLY in Green Boxes

SCCAF Handbook

SCCAF Internal Reference Guide

Lead / Support roles and timing for each stage

Ford PD engineer / Mfg

Manufacturing / Ford PD

SCCAF : Global FMA Website

Model Year & Program Code:

2021 Scorpion 6.7

Part / System Number:

BC3Q-6153-A

Ford Internal Manufacturing Plant:

Rawsonville - USA

Engineering Release Number:

Design Lead:

Ford

Ford Plant GSNB Code:

Part / System Name:

Ring - Oil Control

Ford Customer Plant:

Saarlouis Assembly Plant - Germany

CPSC / COE / UN-UP

050505

VHM3

UN

DFMEA Reference:

2021 B123 - Oil Control

Ford D & R Engineer Approval / Date:

AN Other

AN Other

another@ford.com

08-Mar-23

Ford Manufacturing Engineer Approval / Date:

Ann Engineer

Ann Engineer

anneengineer@ford.com

09-Mar-23

Ford Craftsmanship Approval / Date:

A Part

A Part

apart@ford.com

10-Mar-23

Ford Plant Quality Manager Approval / Date:

Approvals may be obtained by physical signature or electronic sign-off. Recommended: Use DocuSign to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment (See SCCAF FMA webpage for DocuSign help).

Key:

For Special Characteristic definitions refer to Ford FMEA Handbook.

DFMEA : Global FMA Website

From DFMEA:

YC

Sev 9,10 : Potential CC.

YS

Sev 5-8 and may require special control to maintain process capability: Potential SC.

From PFMEA:

CC

Sev 9,10 (Part) : Critical Characteristic.

SC

Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.

OS

Sev 9,10 (Process) : Operator Safety.

HI

Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.

Special Characteristic Totals:

Note 1: Automatic calculation of totals (for guidance only).

Note 2: Count of SC may be less than count of YS.

YC

1

YS

4

CC

1

SC

3

-

1

OS

0

HI

0

Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>.

Special Characteristics Communication and Agreement for all UP content is required by UPV2.

Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.

DFMEA

PFMEA

No.

Characteristic Description

Specification & Tolerance

DFMEA Class

PFMEA Class

Process Control Method

GSPPAS / VPS Sheet #

1

Fixing: Horn Bracket Assembly to Body
Part number: W500212-S442
CPSC-Seq: EE00-130601-734 (Full CPSC-Seq required Letters & Numbers)
Size: M6
Qty: 1
Process: From: JX6T-13801-**- To: JX7B-A16E183-**
Remarks: Horn to side member front assembly LHS

Torque: 10.5 Nm +/- 1.6 Nm
Remarks: N/A

YC

CC

Machine / Device / Fixture:
Electric Tool (DC)
Evaluation Measurement Technique:
Electrical Tool (DC) &
Torque Surveillance as per Global Procedure VOPFAG-241
Control Method:
ISPC - Interlocked to Line & QLS
Frequency:
100% & Torque Surveillance as per Global Procedure VOPFAG-241

FEG00100

2

Fixing: Battery Box Assembly to Battery Trays support
Part number: W719795-S439
CPSC-Seq: EU00-010605-044
Size: M6
Qty: 3
Process: From: JX6T-10723-**- To: JX6T-A16H002-**
Remarks: Battery Box to Battery Tray support on front structure

Torque: 10.5 Nm +/- 1.6 Nm
Remarks: N/A

YS

SC

Machine / Device / Fixture:
Pneumatic Tool
Evaluation Measurement Technique:
Torque Surveillance as per Global Procedure VOPFAG-241
Control Method:
SPC - Torque Control Plan
Frequency:
Torque Surveillance as per Global Procedure VOPFAG-241

FEB00100

3

Electrical Connection: Horn connector properly seated LHS
Harness base part number: 14290
Connector Number: C1RH02-A
Qty: 1
Process: From: 14290 To: 130601
Remarks: N/A

Connection: Socket connection fully engaged
Remarks: Plastic to Plastic Physical Connection

YS

-

PFMEA Occurrence = 3
Machine / Device / Fixture:
Manual Operation

FWE00600

4

Electrical Signal: ABS Sensor Front (LH/RH) Connection
Part number: GN15-2C204-AA
Harness base part number: 14290
Connector Number: C1CA03 & C1CA05
Qty: 2 (1 x LHS / 1 x RHS)
Process: Connect front ABS sensor to front suspension
Remarks: Connect front ABS cable connector to engine bay harness

Signal: No DTCs present in the ABS Module
Remarks: None of the following DTCs present
DTC 0xC0037 - Poor Connection
DTC 0xC003A - Poor Connection
DTC 0x12345 - Description (usually open circuit DTCs)

YS

SC

Machine / Device / Fixture:
EOL Test verification interlocked with QLS
Evaluation Measurement Technique:
ABS Selftest
Control Method:
ABS collect Codes
Frequency:
100%

N/A

FAF03-111-2 : Externally Supplied Content

FAF03-111-2 v2.2 March 2023

Special Characteristics Communication and Agreement - External

OPTIONAL: SCCAF Draft Vers: Date:

DFMEA & Potential Special Characteristics

Type ONLY in Yellow Boxes

PFMEA & Special Characteristics Agreement

Type ONLY in Green Boxes

SCCAF Handbook - Global Phased PPAP - Web Site

See SCCAF Section

SCCAF Handbook : SCCAF Global FMA Website

(Suppliers Only - Accessible through Ford Supplier Portal).

(Ford Internal Only)

Lead / Support roles and timing for each stage

Ford PD engineer / Supplier

Supplier / Ford PD

SCCAF : Global FMA Website

Model Year & Program Code:

2021 B123 MCA

Part / System Number:

BC3Q-6153-A

Supplier Name / Plant:

John Doe Inc. - Pleasantville Plant - USA

Engineering Release Number:

Design Lead:

Ford

Supplier GSNB Code:

A1234

Ford Customer Plant:

Saarlouis Assembly Plant - Germany

Part / System Name:

Ring - Oil Control

CPSC:

050505

DFMEA Reference:

2021 B123 - Oil Control

Ford D & R Engineer Approval / Date:

AN Other

AN Other

another@ford.com

08-Mar-23

Ford STA Engineer Approval / Date:

Ann Engineer

Ann Engineer

anneengineer@ford.com

09-Mar-23

Ford Craftsmanship Approval / Date:

A Part

A Part

apart@johndoeinc.com

10-Mar-23

Supplier Plant Quality Manager Approval / Date:

Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).

Key:

For Special Characteristic definitions refer to Ford FMEA Handbook.

DFMEA : Global FMA Website

(Ford Internal Only)

From DFMEA:

YC

Sev 9,10 : Potential CC.

YS

Sev 5-8 and may require special control to maintain process capability: Potential SC.

From PFMEA:

CC

Sev 9,10 (Part) : Critical Characteristic.

SC

Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.

OS

Sev 9,10 (Process) : Operator Safety.

HI

Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.

Special Characteristic Totals:

Note 1: Automatic calculation of totals (for guidance only).

Note 2: Count of SC may be less than count of YS.

YC

4

YS

6

CC

4

SC

4

-

2

OS

0

HI

0

Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>.

Special Characteristics Communication and Agreement for all UP content is required by UPV2.

Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.

DFMEA

PFMEA

No.

Characteristic Description

Specification & Tolerance

DFMEA Class

PFMEA Class

Process Control Method

1

Sheet Steel Material Properties

WSS-MIA365-A21

YC

CC

Material Certification per batch with material usage controls at manufacturing stations

2

Sheet Steel Material Thickness

1.8 mm Thickness to ESBAU5-1N260-A

YC

CC

Material Certification per batch with material usage controls at manufacturing stations

3

Motor Crank Nut Torque

25Nm ± 2Nm

YC

CC

100% torque control
DC nut runner with Interlock

4

Mounting hole diameter

2.5mm +0.0mm, -0.03mm

YS

-

CMM
PPK=3.45

5

Mounting hole position

X345 Y150 Z-68
Positional tolerance Ø1.6mm

YS

SC

100 % go / no go fixture

6

Bush Push In Force

8kN to 12kN

YS

SC

100% in process with load cell

7

Wire Diameter

Ø 12.7mm ± 0.2mm

YS

-

Vernier Caliper
PFMEA Occurrence = 3

8

Surface Coating (Corrosion Protection)

E-Coat Powder coat as per WSS-M2P177-C2
paint thickness min. 20 µm - max 35 µm

YS

SC

Paint Thickness Ultrasound Tester with SPC

9

Alternator voltage

14.4 - 14.6 V with requested voltage of 14.5V

YS

SC

100% EOL voltage tester

10

Material Properties (Flammability - FMVSS 302)

Carpet Surface : WSS-MBP16-B.
Foam Underlay : WSS-M9P32-D1
Insulator : WSS-M9P32-D1

YC

CC

Material Certification per batch with material usage controls at manufacturing stations

Common naming convention:

- Allows consistency when reporting and assessment.
- Easy recognition of the required SCCAF.

When creating the SCCAF object in FEDE consider including the part and program information. The “Guided Search” function in FEDE does not currently show a column for program. Consistent and detailed naming of “SCCAF Objects” will help all users to locate their require SCCAFs. This will be required for both Internal and External SCCAFs as they will require separate “SCCAF Objects”.

External

- “External SCCAF – program - part number or name – Supplier - Country”
External SCCAF - B123 - A12B_5K952_A – ZhangEng - China

Or

- “External SCCAF – program - part number or name – Supplier – GSDB”
External SCCAF – Trans Oil Cooler - ZhangEng- 6QH4

Internal

- “Internal SCCAF – program - part name/CPSC - Ford Plant”
Internal SCCAF - B123 – Twistbeam – FTM Thailand

Or

- “Internal SCCAF – CPSC - part name – program – plant - version”
Internal SCCAF – 120104 – HVAC Main Unit - B123 – FTM Thailand – vers1

For complex programs with multiple Suppliers and Ford Manufacturing locations consider creating a matrix to understand the number of SCCAFs for your commodities.

This will help understand workload for both LEA D&R Engineers and Quality / Design Assurance teams.

The number of SCCAFs for both External & Internal will need to be identified prior to the end of V0 and recorded on the GPDS FMA Plan.

Ford Manufacturing 1				Ford Manufacturing 2	
Supplier	Supplier	Supplier	Supplier	Supplier	
?	?	?	?	?	EXT
?	?	?	?	?	INT
A123	B567	B567ST	C789	C789	

Ford Manufacturing 3			Ford Manufacturing 4		
Supplier	Supplier	Supplier	Supplier	Supplier	
?	?	?	?	?	EXT
?	?	?	?	?	INT
A123	B567	B567ST	B567	B567ST	

This is where Ford has directed a Tier 1 Supplier to use a specific or a choice from a specific list of Tier 2 Suppliers for a certain component.

The process is detailed in [Directed Tier 2 Sourcing - Global Purchasing Procedure PP – 2.1.2](#)

There are 2 main types of agreement:

- **Traditional Directed Sourcing:** [1]
 - Straight Directed Sourcing: Tier 1 Assembly Supplier has complete quality responsibility for managing and approving Directed Tier 2 Components (including component part approval e.g. PPAP).
 - Basket Directed Sourcing: As above, but Tier 1 assembly Supplier selects Tier 2 Component Supplier from an approved small 'basket' of at least two Ford recommended Suppliers for the Commodity.
- **Triangular Directed Sourcing:**
 - Directed Tier 2 Component Suppliers are responsible directly to Ford for Warranty and financial consequences due to failure of their Components. Tier 1 Assembly Supplier is responsible for Warranty and financial consequences due to failure of only its Assembly.

The responsibility for both DFMEA & SCCAF creation will vary between the 2 main types of agreement.

Below is a table showing responsibilities:

	DMEA		SCCAF - LHS Creation		SCCAF - RHS Creation		SCCAF - Approval	
	Component (Tier 2)	Assembly (Tier 1)	Component (Tier 2)	Assembly (Tier 1)	Component (Tier 2)	Assembly (Tier 1)	Component (Tier 2)	Assembly (Tier 1)
Traditional Directed Sourcing: Straight Directed Sourcing Basket Directed Sourcing	As per agreed Design Responsibilities identified in ESOW		Ford PD (With Supplier Support)		Tier 1		Tier 1 / STA / Ford PD	
Triangular Directed Sourcing:	As per agreed Design Responsibilities identified in ESOW		Ford PD (With Supplier Support)		Tier 2	Tier 1	Tier 2 / STA / Ford PD	Tier 1 / STA / Ford PD

Examples:

- Suspension bushing from a specific Supplier is required into a control arm that is supplied as an assembly.
- Windscreen Wiper Assembly where the Motor Supplier is specifically requested.

[1] FMA strategy is the same as the standard sourcing when a Supplier is Design Responsible.

There are some cases where the straight sourcing (Tier 1 to Ford) relationship is not followed; this can be for both Internal and External SCCAFs.

Suppliers Assembly in Ford Facility.

A Supplier may set up a manufacturing facility within a Ford facility, manufacturing sub-assemblies within the Ford factory and delivering direct to the assembly line.

In this scenario, the External SCCAF is used and should detail the Product Characteristics relating to the assembly process being carried out only. The Product Characteristics relating to the individual components should already be covered by the respective Suppliers External SCCAF covering the manufacture of the part/assembly.

Some examples of this include:

- Wheel and tire assembly.
- Interior console assembly.

Post Assembly Modification

In certain cases, work will be completed on vehicles after the Ford final assembly process and prior to delivery to dealers. This is usually outsourced to a Supplier but can also include the work being undertaken within Ford facilities. The SCCAF form used is dependent on this choice of Ford or Supplier.

The normal SCCAF process must be completed for these scenarios where we detail the Product Characteristics that are required to be controlled during the relevant assembly process.

If work is being undertaken by Ford, then form FAF03-111-1 Internal Ford Manufactured and Assembled Content is used. If work is being undertaken by a Supplier, then form FAF03-111-2 Externally Supplied Content is used.

Some examples of this include:

- Rear Spoiler and wheel arch mouldings are added for a Special Edition vehicle pack.
- Engine component modifications for Hi-Performance variant.

In Production - TVM

If the change is to a new manufacturing site of an existing Supplier, localisation to a new Supplier source or a new Ford Internal Manufacturing location, then a new SCCAF is mandatory for that new location even if it's a carry-over part number. The SCCAF will ensure that the correct controls are in place for each manufacturing location.

For a design change to an existing component or assembly, the Engineering team introducing the change should review the DFMEA and relevant FMA documentation as per Cost Reduction Idea Database (CRID) process using the change point approach.

If the design change identifies new or redundant Potential Special Characteristics (YC / YS), the SCCAF should be updated and signed by all parties as per table on [p8](#).

The engineering release number related to the engineering change must be entered into the "Engineering Release Number" field in the Header.

In Production - Localisation

If the change is to a new manufacturing site of an existing Supplier, localisation to a new Supplier source or a new Ford Internal Manufacturing location, then new SCCAF is mandatory for that new location even if maintaining a carry-over part number.

The SCCAF will ensure that the correct controls are in place for each manufacturing location.

The engineering release number related to the engineering change must be entered into the "Engineering Release Number" field in the Header.

External

1	Model Year & Program Code:	2021 B123 MCA	6	Part / System Number:	BC3Q-6153-A
2	Supplier Name / Plant:	John Doe Inc. - Pleasantville Plant - USA	7	Engineering Release Number:	
3	Supplier GSDB Code:	A1234	8	Design Lead:	Ford
4	Part / System Name:	Ring - Oil Control	9	Ford Customer Plant:	Saarlouis Assembly Plant - Germany
5	CPSC:	050505	10	DFMEA Reference:	Generic DFMEA - Oil Control - 050500

1. Enter the Model Year and Program Code of the Vehicle or Powertrain program for this part or family of parts. e.g. 2018 B562MCA.
2. Name of the specific Supplier and the manufacturing plant location (City and Country) e.g. Klein Stamping - Cologne – Germany.
3. Enter the 5-character GSDB code representing the External Supplier Manufacturing Site for this part.
4. Detailed Part / Assembly or System name.
5. Enter the CPSC code relating to the commodity; this must align with the CPSC in the FMA Plan & Sign off Form. (Global FMA Tracker if used).
6. Include Prefix, Base Part Number and Variant (first letter of Suffix) for all parts that the SCCAF controls.
7. Leave blank for not yet released components in a GPDS program. Include WERS release number for ongoing production SCCAFs.
8. Identify whether the Design Lead (DFMEA owner relating to SCCAF) is Ford or Supplier (select from Dropdown). DFMEA responsibility should align with Global FMA Tracker or FMA Plan and Sign Off Form.
9. Location of the Ford or JV Final Manufacturing Plant where this Component/Assembly will be assembled e.g. Chennai, India / Louisville, USA / Valencia, Spain / Pacheco, Argentina.
10. Enter the details of the DFMEA that the SCCAF is based on (refer to Design Lead).
 - **Ford** : Enter FEDE MY & Program Code and DFMEA name or Generic (if based off Generic DFMEA).
 - 2021 B123 Oil Control.
 - Generic DFMEA - Oil Control 050500.
 - **Supplier** : Enter Unique Supplier DFMEA identification.
 - John Doe - DFMEA 1234-9876.

Internal

1	Model Year & Program Code:	2021 Scorpion 6.7			6	Part / System Number:	BC3Q-6153-A
2	Ford Internal Manufacturing Plant:	Rawsonville - USA			7	Engineering Release Number:	
3	Ford Plant GSDB Code:	-			8	Design Lead:	Ford
4	Part / System Name:	Ring - Oil Control			9	Ford Customer Plant:	Saarlouis Assembly Plant - Germany
5	CPSC / COE / UN-UP	050505	VHM3	UN	10	DFMEA Reference:	Generic DFMEA - Oil Control - 050500

As per External with the following exceptions:

Note: For VOME (Ford Assembly Plant) SCCAFs, items 2 and 3 can be left blank.

2. Name of the specific Ford Internal Manufacturing Plant and its location (City and Country). e.g. Klein Stamping - Cologne – Germany.
3. Enter the 5-character GSDB code representing the Ford Plant Manufacturing Site for this part.
5. Include the CPSC as per External SCCAF above, the relevant PMT and Workstream either UN or UP.

Note: When using FEDE FMA Plan created SCCAFs: the header will auto populate from the information entered and viewed through the “Overview” screen. No updates can be completed within the SCCAF template and must be edited in the “Overview” screen. FEDE will then overwrite the header information on the next SCCAF download.

Multiple Parts on External SCCAFs

[Contents Page](#)

Separate SCCAFs are required if there are many differing Product Characteristics to control between multiple parts. If there are only a few minor differences in Product Characteristics requiring control, then multiple parts may be contained within one SCCAF for efficiency.

A general rule is “One Drawing = One SCCAF” assuming there is only a single manufacturing location. An individual SCCAF will be needed for all manufacturing locations on a make to print drawing.

If multiple parts are shown on a single SCCAF then clear differentiation must be detailed on the SCCAF, to show which characteristics vary between the different parts and the individual values for each part.

This allows each part to be tracked and measured through the APQP/PPAP Evidence Workbook and also help avoid confusion during manufacturing.

This can be either showing the individual part numbers on the SCCAF itself (Example 1) or by referring to a separate tab (Example 2) within the SCCAF template detailing the part numbers, the characteristics and dimensional values for each part included on the SCCAF.

Some examples:

- Trim Parts with different colours or grain types.
- Hoses with varying lengths.
- Springs with varying load rates.
- Switches with different graphics or colours.

Example 1

Model Year & Program Code: 2021 B123 MCA	Part / System Number: AB3C-1234-A / B / C
Supplier Name / Plant: John Doe Inc. - Pleasantville Plant - USA	Engineering Release Number:
Supplier GSDB Code: A1234	Design Lead: Ford
Part / System Name: Front Coil Spring	Ford Customer Plant: Saarlouis Assembly Plant - Germany
CPSC: 040101	DFMEA Reference: Generic DFMEA - Front Road Spring - 040101

Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA - Global FMA Website (Ford Internal Only)	From DFMEA: <table border="1"> <tr> <td>YC</td> <td>Sev 9,10 : Potential CC.</td> </tr> <tr> <td>YS</td> <td>Sev 5-8 and may require special control to maintain process capability : Potential SC.</td> </tr> </table>	YC	Sev 9,10 : Potential CC.	YS	Sev 5-8 and may require special control to maintain process capability : Potential SC.	From PFMEA: <table border="1"> <tr> <td>CC</td> <td>Sev 9,10 (Part) : Critical Characteristic.</td> </tr> <tr> <td>SC</td> <td>Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.</td> </tr> </table>	CC	Sev 9,10 (Part) : Critical Characteristic.	SC	Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.		
YC	Sev 9,10 : Potential CC.											
YS	Sev 5-8 and may require special control to maintain process capability : Potential SC.											
CC	Sev 9,10 (Part) : Critical Characteristic.											
SC	Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.											
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS	<table border="1"> <tr> <td>YC</td> <td>YS</td> </tr> <tr> <td>0</td> <td>8</td> </tr> </table>	YC	YS	0	8	<table border="1"> <tr> <td>CC</td> <td>SC</td> <td>-</td> </tr> <tr> <td>0</td> <td>8</td> <td>0</td> </tr> </table>	CC	SC	-	0	8	0
YC	YS											
0	8											
CC	SC	-										
0	8	0										

DFMEA			
No.	Characteristic Description	Specification & Tolerance	DFMEA Class
1	Load: AB3C-1234-A	3680N ± 3%	YS
1	Load: AB3C-1234-B	3820N ± 3%	YS
1	Load: AB3C-1234-C	3950N ± 3%	YS
2	Wire Diameter: AB3C-1234-A	Ø 11.9 mm ± 0.1mm	YS
2	Wire Diameter: AB3C-1234-B	Ø 12.4 mm ± 0.1mm	YS
2	Wire Diameter: AB3C-1234-C	Ø 13.2 mm ± 0.1mm	YS
3	Material Specification	SAE J404 9254	YS
4	Inner Diameter	Ø 82mm ± 1.5mm	YS

Example 2

DFMEA			
No.	Characteristic Description	Specification & Tolerance	DFMEA Class
1	Load	See Part Info Tab	YS
2	Wire Diameter	See Part Info Tab	YS
3	Material Specification	SAE J404 9254	YS
4	Inner Diameter	Ø 82mm ± 1.5mm	YS

	A	B	C
	Part Number	Load ± 3%	Wire Diameter ± 0.1mm
1	AB3C-1234-A	3680N	Ø 11.9 mm
2	AB3C-1234-B	3820N	Ø 12.4 mm
3	AB3C-1234-C	3950N	Ø 13.2 mm
4			
5			

The Product Characteristics must be taken from the DFMEA and must be specific, comprehensive and cover **all** the failure modes identified within the DFMEA.

Failure Modes or Causes cannot be simply cut and pasted from the DFMEA and will often require further refinement to define the appropriate Design Characteristics.

If the DFMEA is not to an adequate standard, the team can add extra Product Characteristics from prior engineering knowledge that have not been included. The program and Core Global DFMEA should then be updated to include the missing Product Characteristics.

Product Characteristics can only be modified / deleted by agreement with LEA D&R Engineer, changes requested by other team members (Ford Manufacturing, STA, Supplier, etc.) must be first agreed upon by the LEA D&R Engineer.

- A Product Characteristic is a feature that must be maintained to ensure intended product function (not manufacturing errors).
- Some examples include dimensions, size, shape, location, orientation, texture, hardness, tensile strength, appearance, coating, or reflectivity.
- A Product Characteristic should be published on a drawing or in a specification document. If it is not, then it is not a properly defined Product Characteristic.
- A Product Characteristic can be measured.
- A Product Characteristic may identify features that affect the customer or regulatory compliance.

All Product Characteristics must be:

Specific and measurable (may be attribute e.g. pass/fail or variable e.g. length in mm).

AND Controllable on an ongoing basis.

AND Affected by and specific to the manufacturing process covered by the specific SCCAF
(see YS definition [p24](#)).

The following examples **should not** be included in the Product Characteristic column on the SCCAF document, some of these items may be communicated using the “Manufacturing Info Transfer Sheet” for more information see [p22](#):

- DV / PV test requirements, durability requirements, and corrosion test requirements. These are not typically measurable or controllable in ongoing production.
- Requirement names i.e. SU-1014. The actual Product Characteristics that result in compliance with the requirement should be defined to determine what must be controlled.
- Part Presence such as number of hardware. This is a process error and should not have DFMEA class. Part presence should be analysed in the PFMEA.
- Measurements that are required for additional information but are not a YS.
- Manufacturing errors such as “Do not drop”, “No Pinch, Kink or Damage from Installation should not be included in the SCCAF as they would not be identified in the DFMEA therefore not YS or YC. These should be reviewed through the PFMEA and Process Control Plans.
- Periodic tests: the SCCAF is for ongoing Production Controls, so should not include monthly, 6 monthly or yearly tests. The controls and reaction plans for the periodic tests should be specified in the ES and included in the Control Plans.
 - An example is “Flammability (FMVSS 302)” this is usually a yearly test and therefore would **not** be included in the SCCAF. Ongoing controls of Material Properties would be included in the SCCAF and this will ensure compliance to regulations through all of production with confirmation testing at required intervals.

Note: There must be no strikethrough items on a SCCAF; if the line item is not required, it should be removed. The SCCAF is the document to cascade Special Characteristics. The SCCAF document is imported into the APQP/PPAP Evidence workbook to ensure that all characteristics are carried through and documented. If a SCCAF has characteristics that are strikethrough these will be imported without the strikethrough and hence showing a larger amount of characteristics than agreed in the SCCAF template.

Note: Additional items may be added on to the SCCAF that have been driven out of the PFMEA review. These items do not require a DFMEA classification as they are driven from the PFMEA and are process related and controls may necessitate the request of tools/gauging.

Ford Owned DFMEA

For Ford owned DFMEAs, Product Characteristics are defined in the “Product Characteristics” column within the “Prevention” section.

Function Description			Failure Mode				Current Design Controls									
Input/ Assumptions	Verb / Noun	Desired Output/ Engineering Measurable	Potential Failure Mode	Potential Effects	S E V	C l a s s	Potential Cause(s)/ Mechanism(s) of Failure	Prevention		O C C	Detection and/or Verification (Pre-<xDJ>)		Detection Validation (Post-<xDJ>)		D E T	R P N
								Design Methods	Product Characteristics		Virtual Check	Physical Test	Virtual Analysis	Physical Test		

Supplier Owned DFMEA

Review DFMEA and identify Product Characteristics from:

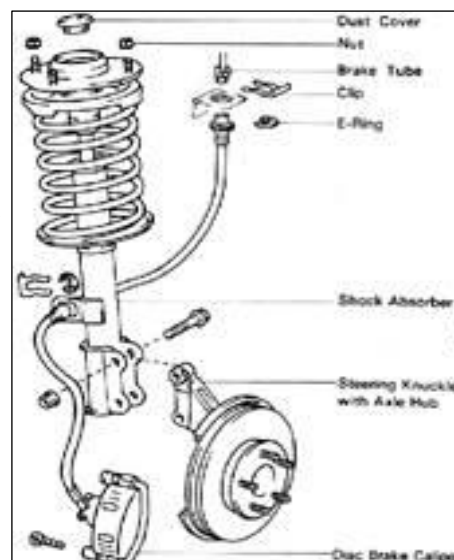
- Potential Cause/Mechanism of Failure.
- Countermeasures identified in Design Controls – Prevention.

Translate the Potential Cause Of Failure into the Design Characteristic that manufacturing is required to control to ensure the effect of failure cannot occur.

Detail this engineering Design Characteristic into the “Characteristic Description” column in the SCCAF.

Example shown below for Knuckle to Shock Absorber interface.

1. Material properties.
2. Clevis width.
3. Bolt tensile class.
4. Bolt torque.



Item/Function	Potential Failure Mode	Potential Effect(s) of Failure	S e v	C l a s s	Potential Cause/ Mechanism of Failure	Current Design Controls	
						Prevention Controls	O C C
[2] 2. Transfer Loads Dynamic 2.1 Transfer loads to/from Shock Absorber axis Pt. 7 / Pt. 36 2.1.1 Transfer loads through Strut Tube Clevis Joint Pt. 36 (Option A) ±x / [kN] ; ±y / [kN] ; ±z / [kN] ; 240,000 km / 10 YIS	[2.1] No load transfer through Strut Tube Clevis	Knuckle: - Knuckle Assembly: - Front Suspension: Inoperable (8) Vehicle: Inoperable (8) Customer: loss of vehicle control (10) Government Regulations: -	10	YC	[2.1.1] Wrong Material specified 1	- CAE calculation - Material according to ESACPI-3K171-** - Specification: SU-1012 Durability Endurance	1
				YC	[2.1.2] Clevis thickness wrong dimensioned 2	- CAE calculation - Tolerance stack up - Knuckle Design Rule: The strut clevis bracket interface must be aligned to the strut tolerances.	1
				YC	[20.1.9] Bolt property class too low 3	- CAE calculation - Joint calculation acc. to SU 1014	1
				YC	[20.1.10] Bolt tightening torque wrong specified 4	- CAE calculation - Joint calculation acc. to SU 1014	1
				YC	[20.1.11] Knuckle cleanliness wrong specified	Knuckle Design Rule: The Cast Knuckle Drawing must be according to Guide File Drawing GUACPI-3K171-**: All Views, Sections, Datum, GD&T and Text on the knuckle drawing must have the same position and call out on the application drawings.	2

DA/AIAG FMEA Handbook release (June 2019)

- AIAG & VDA combined their individual FMEA Handbooks into a single, AIAG/VDA Handbook v1.0.
- Where Suppliers are making use of the new FMEA approach, Ford will accept the new format.
- Until the revised Ford Motor FMEA Handbook is published, the current FMEA Handbook 4.2 is the master reference For more information see [FMA Webpage – DFMEA](#).

Flammability:

- FMVSS302 or equivalent flammability requirements are usually certified annually. We need to make sure that all product manufactured will meet the requirement & not just the yearly certified components.
- Therefore, we control the Material Properties of the components ongoing, if the components are made of the correct material and this will ensure compliance to regulations through all of production with confirmation testing at required intervals (see example [p46](#)).

Part Presence:

- Part Presence should not be included in the SCCAF, as the SCCAF is not the BOM. The DFMEA assumes that the part is made correctly and therefore would not drive a YS.
- It is important to analyse this in the PFMEA for all ways that the part may not be included in the assembly.
- The new Manufacturing Info Transfer Sheet can be used to identify part presence if the team wish to make sure this information is specifically cascaded to the Supplier or Internal Manufacturing.
- Often a check on a specific characteristic will also confirm part presence e.g. If there is a bolt torque specified this will also check that the presence of the bolt.

Part Marking & E8 Branding:

- This will not normally be included on a SCCAF as for most manufacturing process (e.g. casting and moulding) the E8 Branding and other part information will be included in the tooling.
- Therefore they will not vary enough during manufacturing of the component and will not meet the criteria for “influenced by manufacturing” to become a YS (see [p24](#) & [p25](#) for info on YS).
- For components with separate labels that are attached to a component these will require ongoing controls and verification.

Recycling – Materials & Marking

- Material recyclability will be controlled by using the correct material approved at the start of the program and this will be controlled by the “Material Certificate”. An individual line item for “Material Recyclability” is not required, if the “Material Properties” are already included in the SCCAF.
- Recycling labelling requirements on the part are dependent on the DFMEA class and whether they are included in the tooling or as a separate label. This outcome will vary around local government regulations and determine whether it’s a YC. Teams will need to analyse each label for their individual designs the information below is provided as general guidance only.
 - YC – In Tool : This can be included in the SCCAF but as it will be cut in the tooling the team can use similar wording to the example shown in “example 13 - Head Impact Radius” on [p44](#).
 - YC – Label : For components with separate labels that are attached to a component these will require ongoing controls and verification that all labels are included and ensure that non-conforming product is not shipped (e.g. vision system to confirm label presence).
 - YS – In tool : Would not be included on the SCCAF as it is in the tooling and therefore not influenced by manufacture and would not become a YS in the DFMEA (see [p25](#)).
 - YS – Label : For components with separate labels that are attached to a component these will require ongoing controls and verification (e.g. vision system to confirm label presence).

Standard Fasteners:

- A SCCAF is not required for Global Fasteners (W5, W7) parts, there is a detailed procedure and process for their release.
- If the fastener has a program specific part number (prefix-base-suffix) or W9 fastener then a SCCAF would be required to meet the FAP requirement of all NTEIs for a program.

Appearance Approval Request/Master Samples:

- Appearance item defects (sink marks, flash etc.) should technically NOT be included on a SCCAF but rather the controls relating to appearance outcomes should be identified through the PFMEA and included on the Suppliers production control plans.
- If appearance approval items are included on the SCCAF they should, where possible, be specific and measurable. For example, if the specific appearance characteristic is meeting "colour master", then an appropriate control would be to use a colour spectrometer measuring 3 pieces per run (start, middle and end with SPC) or if the characteristic relates to "gloss" than the use of a gloss meter to determine the gloss value is within specification per run with SPC is acceptable.
- Where a specific measurable is not possible a visual check to a master sample AAR is allowable provided that appropriate KPIVs are identified on the SCCAF with appropriate controls.
- Suppliers of decorative components (Interior, Exterior and Underhood) must comply with the GDCAP (Global Decorative Approval Process) document. This document outlines all the necessary steps to achieve appearance approval for PPAP.
- The Paint, Colour Control Guideline documents outline the best practice colour control processes which should be used by the Supplier to achieve on-going production quality in line with the approved appearance approved sample (master sample).

These can be accessed on the Global Phased PPAP website under “Ford Colour Harmony”.

(https://web.qpr.ford.com/sta/Phased_PPAP.html).

Manufacturing Information Transfer Sheet

[Contents Page](#)

This optional sheet is available through:

- “PFMEA Info Transfer Sheet” tab in FAF03-111-2 Externally Supplied Content.
- “Manuf Info Transfer Sheet” tab in FAF03-111-1 Internal Ford Manufactured & Assembled Content.

This is an **optional** form that can be used by D&R Engineers to cascade non YC-YS information for consideration by manufacturing. PD chooses to initiate the form only when non YC-YS information is identified. To initiate the form, the D&R engineer enters the additional information in the LHS of the form. Once PD has initiated the form by completing the LHS then a “Manufacturing Response” (RHS) is required from the Manufacturing team (Internal or External) to complete the SCCAF Process.

Use this tab to transfer any additional information to the Supplier / Ford Manufacturing that the engineering teams want to be considered during the Product Process development including PFMEA / Process Control Plan / Work Instructions that the team believe is not adequately communicated by existing documents. This information may come from many sources including but not limited to Quality History or periodic test requirements.

This sheet **must not** be used for YS or YC items identified through the DFMEA.

It is intended to communicate important characteristics (not YS/YC) or process related information that the design team wants to ensure the manufacturing process team (Supplier / Ford Manufacturing) has considered associated process controls in the process related documentation such as PFMEA and Process Control Plan.

Information that can be transferred using the sheet (provided it is not a YC or YS) includes but not limited to:

- Part presence including hardware.
- Avoidance of manufacturing errors such as “Do not drop” & “Correct routing” etc.
- Periodic tests such as monthly, 6 monthly or yearly In Process (IP) tests.
- Tests that are recommended or required at EOL only to detect manufacturing errors, such as shaker tables for loose component in Instrument panel or centre console.

It is not necessary to duplicate information, where there are existing Ford Procedures to transfer information to Supplier / Ford Manufacturing such as Engineering Specifications for IP (In-Process) testing that are referenced on released data (CAD model, 2D drawing). But if deemed useful the team may choose to use this sheet to reference the information.

V1 / Unit PTC: LEA D&R Engineer:

1. Determines if PFMEA Information Sheet is to be used.
- 1a. Completes the Information column with required information.
- 1b. Identifies the severity of the Failure mode relating to this item (External Only).

V2 / Unit TPVDJ: Supplier / Ford Manufacturing facility review the request:

1. Accept the information or discuss with LEA D&R Engineer the reason for non-acceptance.
- 2a. Detail the Process steps put in place to prevent the issue.
- 2b. Confirm whether this is included in the relevant PFMEA and Process Control Plan (External Only).

FAF03-111-2 External SCCAF shown below

[Completed External Example](#)

[Completed Internal Example](#)

Manufacturing PFMEA Information Transfer Sheet (Optional Form) For additional consideration of Non YS or YC characteristics.				
Model Year & Program Name: 2021 B123 MCA Part / System Name: Ring - Oil Control Part / System Number: BC3Q-6153-A Ford Customer Plant: Saarlouis Assembly Plant - Germany		Supplier Name / Plant: John Doe Inc. - Pleasantville Plant - USA Supplier GSDB Code: A1234 Ford D & R Engineer: STA Engineer: Supplier Plant Quality Manager:		
PD optionally initiates the process where non YC-YS information for consideration by manufacturing is identified by entering the additional information in the LHS. Once initiated by PD the full form must be completed with Manufacturing response (RHS) to complete SCCAF Process (see p22 SCCAF Handbook).				
No.	Required information to be transferred to PFMEA Excluding YS or YC Design Characteristics. (Lessons learnt / Quality History)	Failure Mode Severity	Manufacturing Response	Included in PFMEA
1	Scrap if dropped or damaged Parts must be identified to meet all quality standards and specifications, and parts must be undamaged. Damaged parts must be scrapped, no matter what the cause: shipping, dunnage, handling (dropped), etc. Assume dropped parts are damaged.	8	Included in PFMEA and work instructions with appropriate controls	Yes
2	Filler pipe to fuel cap interface leak tightness No Surface Damage to Pipe or Cap Sealing Surface During Handling which would result in detectable leak National Standard Nozzle required during Initial filling	10	Fuel filler equipment fitted with proper nozzle. Preventative Maintenance plan in place to ensure proper nozzle condition.	Yes
3	Loose fastener/items component trapped inside assembly.	4	Clips: Visual inspection. Clips marked up with China marker. Frequency/ Sample size: 100% Turn trunion 180° to check for loose fasteners trapped inside assembly Frequency/ Sample size: 100%	Yes
4	SCR Lines installation routing and handling Installation routing per illustration. No pinching or kinking of SCR lines during installation and handling	7	Included in PFMEA and work instructions with appropriate controls	Yes

Product Characteristic Specification Tolerances: Define the range for each Product Characteristic that must be maintained to ensure intended product function.

Every Product Characteristic identified on the SCCAF must have a corresponding Specification and Tolerance completed with the measurable units e.g. mm, Nm, Amps.

The reason for including Specification & Tolerance in the SCCAF is to ensure compatibility between the design and the process and ensure that the process can both produce and control to the required tolerance level. The CAD model/Drawing is **always** master and the information shown in the SCCAF is to allow the Supplier / Ford Internal Manufacturing to understand early in the design process the type of measurement method required as SCCAF is signed before CAD Model / Drawing completion at xDJ.

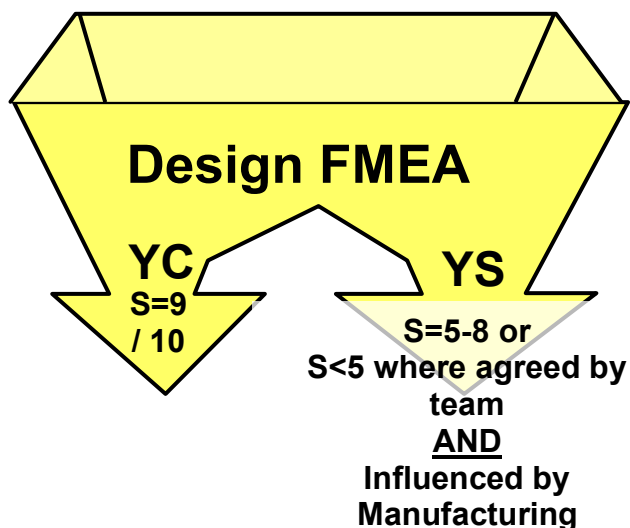
These must be measurable and written in engineering terms. Listed below are some guidelines for completing the "Specification & Tolerance" column:

- Specification of "According to Drawing" is generally not acceptable. An example of an exclusion may be, referencing a table of multiple shock absorber rates on the one shock absorber drawing where the information cannot easily be contained within the SCCAF. The referenced item must be specific and easily understood. For example, the reference should state the section detail, view or table to refer to within the specified drawing.
- For complex dimensions (e.g. surfaces), the Specification may refer to the CAD /Drawing but the tolerance such as the "profile tolerance" should be included in the SCCAF.
- List the actual measurement and spec if possible as it aids the downstream process when importing into the 'APQP / PPAP Evidence Workbook' and Supplier / Ford Manufacturing Process Control Plans.
- When referring to a Specification you need to detail the actual specification number and which section of specification is relevant.
 - ES-AR79-5K009-AA is not acceptable. Use ES-AR79-5K009-AA Section 4.1.
 - VSEM (Vehicle Software Electrical Management) detailed to ROIN # and version.
- All references must be:
 - A WERS released document (e.g. Engineering Spec, Material Spec, Table, CAD, Drawing).
 - An industry Standard (e.g. SAE Standards, DIN Standards).
 - Information attached directly to the SCCAF.
- Ford documents that are not WERS released and therefore are UNACCEPTABLE as references include but are not limited to:
 - SDS Requirements
 - CETP's
 - Design Rules
 - PD Health Charts

Note: Items above can be entered into the Notes Column for reference or under the specific characteristic in the Product Characteristics column in parenthesis e.g. (DR-010807-711906)

- For External SCCAF : Suppliers may withhold proprietary specification and tolerances but Product Characteristics are still to be identified. Add the note "*Supplier Proprietary Information*" and list the Supplier drawing number or Specification document in which the specification and tolerance is shown. This is to show that the SCCAF is not just missing the required information and ensures the source of the information is identified and cascaded to the relevant Supplier documents.
- If the exact value is not available at V1 / Unit PTC, an estimated or surrogate value should be identified. To properly assess compatibility of the process design, the approximate values of the specification and tolerance needs to be known e.g. it is quite different to produce and control to $12.0 \pm 0.1\text{mm}$ vs $12.00 \pm 0.01\text{mm}$.
- A correctly written specification and tolerance should not include the following words:
 - Prevent, Avoid, Ensure, Meet, Comply, No, Maintain.
- If the Spec & Tolerance includes more than one unit of measurement or differing type of measurement (e.g. Position and Parallelism), these should be separated into 2 line items as they will usually require different Process Control Methods e.g.
 - Crimp width: xx +/-xx mm
 - Terminal crimp pull out force: min XX N

Note: Torque and angle are acceptable as these together ensure the bolt is properly torqued.
- Including 2 different dimensions in the Spec & Tolerance will cause issues downstream when the SCCAF is imported into the APQP/PPAP Evidence workbook. The supplier will not be able to complete the necessary information for both dimensions in a single row. Each dimension MUST be separated into two separate SCCAF line items.



Effect	Criteria: Severity of Effect	Ranking
Failure to Meet Safety and/or Regulatory Requirements	Potential failure mode affects safe vehicle operation and/or involves noncompliance with government regulation without warning .	10
	Potential failure mode affects safe vehicle operation and/or involves noncompliance with government regulation with warning .	9
Loss or Degradation of Primary Function	Loss of primary function (vehicle inoperable, does not affect safe vehicle operation).	8
	Degradation of primary function (vehicle operable, but at reduced level of performance).	7
Loss or Degradation of Secondary Function	Loss of secondary function (vehicle operable, but comfort / convenience functions inoperable).	6
	Degradation of secondary function (vehicle operable, but comfort / convenience functions at reduced level of performance).	5
Annoyance	Appearance or Audible Noise, vehicle operable, item does not conform and noticed by most customers (> 75%).	4
	Appearance or Audible Noise, vehicle operable, item does not conform and noticed by many customers (50%).	3
	Appearance or Audible Noise, vehicle operable, item does not conform and noticed by discriminating customers (< 25%).	2
No effect	No discernible effect.	1

Severity is the rank associated with the most serious effect from the Potential Effects column. A reduction in Severity ranking index can be effected only through a design change.

POTENTIAL CRITICAL CHARACTERISTIC (YC):

A Product Characteristic designated in the DFMEA with the YC symbol in the Classification column. These characteristics have a causal relationship to Potential Failure Modes having Severity of Effects rated 9-10.

POTENTIAL SIGNIFICANT CHARACTERISTIC (YS):

A Product Characteristic designated in the DFMEA with the YS symbol in the Classification column, when both the following criteria are met:

1. Characteristic has a causal relationship to Potential Failure Modes having Severity of Effects rated 5-8, or where agreed by the cross-functional team, having Severity of Effects rated <5.
 2. Characteristic may be influenced by the manufacturing process and may require Special Controls to maintain the required process capability.
- **Severity** is the rank associated with the most serious effect from the Potential Effects column.
 - A reduction in Severity ranking index can be effected only through a design change.
 - Severity should be estimated using the table above.

YS for Severity <5

DFMEA analysis of a new A-pillar trim design identifies a failure mode with an effect of a squeak and rattle.

- Per the severity rating table, this is rated as a severity of 3 (Squeak and rattle noticed by 50% of customers).

The team (PD Engineering, Ford Manufacturing, Supplier, STA etc.) agrees that through the Quality History analysis, squeaks from this part have generated significant NPS and Customer Satisfaction loss.

- The team agrees to make the Product Characteristics associated with this failure mode a YS, despite the severity 3 rating. This is documented in the DFMEA as YS and therefore is transferred to the SCCAF.

Not all severity 5-8 will become a YS as they must also be influenced by the manufacturing process. The Engineering team creating the DFMEA needs to consider the manufacturing process when analysing whether a severity 5-8 becomes a YS.

Example 1: Cause is *“Mounting hole is specified too small.”* The component is manufactured using injection moulding and the variation in the dimension to cause the issue is greater than the amount of variation in the process.

In an applied example: The nominal hole size is 10mm but to cause an issue the hole must be smaller than 8mm. Standard injection moulding process does not have sufficient process variation to cause the issue. Therefore, the Product Characteristic is not influenced by manufacturing and does not meet **both** required criteria for a YS.

The severity will remain as 5-8 but no YS will align to this in the DFMEA and therefore is not transferred to the SCCAF.

Example 2: Cause is *“corrosion due to selected material not robust to corrosive environment”* and the Prevention Control is to make the component out of plastic thus avoiding the cause. The Product Characteristic identified will be *“Material Selection - Plastic”* but no matter what plastic material we use in manufacturing, we cannot make the part susceptible to corrosion. A process for making parts in plastic cannot accidentally make parts in metal.

Therefore, the Product Characteristic is not influenced by manufacturing and does not meet **both** required criteria for a YS.

The severity will remain as 5-8 but no YS will align to this in the DFMEA and therefore is not transferred to the SCCAF.

Note: The correct plastic material may be identified as a YS elsewhere in the DFMEA for other reasons.

Some other examples may include but are not limited to:

- **Incorrect Part identification:** The identification marking is cut into the tool and will not vary enough during manufacturing of the component to achieve the failure mode. This therefore is not influenced by manufacturing so does not meet **both** required criteria for a YS.
- **Rib Thickness on plastic components:** Thickness is required for strength but due to the manufacturing process used will never vary enough to cause the failure mode. Therefore this Product Characteristic is not influenced by manufacturing so does not meet **both** required criteria for a YS.
- **Clip tower dimensions:** The dimensions and locations are important to support the clips but due to the manufacturing process used will never vary enough to cause the failure mode. Therefore this characteristic is not influenced by manufacturing so does not meet **both** required criteria for a YS.

At V1 / Unit PTC, all Product Characteristics must align to a DFMEA class.

At V2 / Unit TPVDJ, a SC Product Characteristic can be fed back from Ford Manufacturing or the Supplier relating to the process but **must** have a blank DFMEA class.

For more information on FMA GPDS Deliverables including SCCAF : [FMA Deliverables Table - GPDS Gateways](#).

For more details on how to store SCCAFs in FEDE when using **Global FMA Tracker** : [SCCAF Storage](#) document on [Global FMA Website - SCCAF](#) or [FEDE Help SPLs](#).

The SCCAF deliverable is complete for V1 / Unit PTC when:

1. All Product Characteristics have been transferred from the program specific DFMEA, with a Specification and Tolerance that is measurable and DFMEA classification complete (YC or YS).

Special Characteristics Communication and Agreement - External				OPTIONAL:	SCCAF Draft Vers:	Date:
Lead / Support roles and timing for each stage	Ford PD engineer / Supplier	Supplier / Ford PD	SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section SCCAF Handbook - SCCAF Global FMA Website			
Model Year & Program Code:	2021 B123 MCA	Part / System Number:	BC3Q-6153-A			
Supplier Name / Plant:	John Doe Inc. - Pleasantville Plant - USA	Engineering Release Number:				
Supplier GSDB Code:	A1234	Design Lead:	Ford			
Part / System Name:	Ring - Oil Control	Ford Customer Plant:	Saarlouis Assembly Plant - Germany			
CPSC:	050505	DFMEA Reference:	2021 B123 - Oil Control			
Special Characteristics for: 2021 B123 MCA John Doe Inc. - Pleasantville Plant - USA BC3Q-6153-A						
Approval Signature		Name		e-mail Address		Approval Date
Ford D & R Engineer						
Approval / Date:						
Ford STA Engineer						
Approval / Date:						
Ford Craftmanship						
Approval / Date:						
Supplier Plant Quality Manager						
Approval / Date:						
Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).						
Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA - Global FMA Website (Ford Internal Only)						
From DFMEA: YC Sev 9,10 : Potential CC. YS Sev 5-8 and may require special control to maintain process capability: Potential SC.		From PFMEA: CC Sev 9,10 (Part) : Critical Characteristic. SC Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.		OS Sev 9,10 (Process) : Operator Safety. H Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.		
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS.		YC: 4, YS: 6, CC: 0, SC: 0		OS: 0, H: 0		
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>. Special Characteristics Communication and Agreement for all LP content is required by LPV2. Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.						
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC			
2	Sheet Steel Material Thickness	1.8 mm Thickness to ESBAU5-1N260-A	YC			
3	Motor Crank Nut Torque	25Nm ± 2Nm	YC			
4	Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS			
5	Mounting hole position	X345 Y150 Z:68 Positional tolerance Ø1.6mm	YS			
6	Bush Push In Force	8kN to 12kN	YS			
7	Wire Diameter	Ø 12.7mm ± 0.2mm	YS			
8	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	YS			
9	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS			
10	Material Properties (Flammability - FMVSS 302)	Carpet Surface : WSS-M8P16-B. Foam Underlay : WSS-M99P32-D1 Insulator : WSS-M99P32-D1	YC			

2. SCCAF Assessment – V1 tab (External Only) completed and shows 100% Complete.

V1 External SCCAF Assessment for 2018 B123 MCA - Ring - Oil Control - John Doe Inc. - Pleasantville Plant - USA			
Current Status:	PASS	Milestone Report Status:	COMPLETE
Assessor:	Graham Monks	Date:	7-Oct-2018
Notes on Assessment			Assessment Completion
Exit Full Screen			Completion % 100%
			20 Questions Answered out of 20 total. 0 Questions remain to be answered.

3. SCCAF has been cascaded to the Suppliers (External) / Ford Internal Manufacturing (Internal).
 - a. SCCAF has been cascaded to the relevant STA Site Engineer (if required) for all External SCCAFs.
4. "SCCAF Object" created in FEDE for Each Individual SCCAF.
5. SCCAF with completed LHS (Internal & External) has been stored in the "Edit SCCAF" screen of its specific "SCCAF Object" in FEDE, do not store to "Attachments" screen.
6. SCCAF Assessment for each External SCCAF has been loaded to "Attachments" screen in FEDE.
7. All SCCAFs have been submitted in the "Request Approval" screen in FEDE this will create a "Published" and locked version of the V1 / Unit PTC SCCAF Documents.
8. All SCCAFs (Internal & External) have been marked as "LHS Complete – Loaded to FEDE" and "Cascaded" in "Global FMA Tracker".

SCCAF	Required	LHS Complete Loaded to FEDE	Cascaded
Internal SCCAFs	1	1	1
External SCCAFs	2	2	2

For help see [SCCAF Storage](#) document.

For more information on FMA GPDS Deliverables including SCCAF : [FMA Deliverables Table - GPDS Gateways](#).

For more details on how to store SCCAFs in FEDE when using **FMA Plan in FEDE** : [SCCAF Completion & Approval](#) document on [Global FMA Website - FMA Plan](#) or [FEDE Help SPLs](#) - [FEDE Help-SCCAF Enhanced Workflow Process](#).

The SCCAF deliverable is complete for V1 / Unit PTC when:

1. All Product Characteristics have been transferred from the program specific DFMEA, with a Specification and Tolerance that is measurable and DFMEA classification complete (YC or YS).

Special Characteristics Communication and Agreement - External				OPTIONAL: SCCAF Draft Vers.	Date:
Lead / Support roles and timing for each stage	Ford PD engineer / Supplier	Supplier / Ford PD	SCCAF Handbook - Global Phased PIPAP - Web Site See SCCAF Section SCCAF Handbook - SCCAF Global FMA Website		
Model Year & Program Code:	2021 B123 MCA	Part / System Number:	BC3Q-6153-A		
Supplier Name / Plant:	John Doe Inc. - Pleasantville Plant - USA	Engineering Release Number:			
Supplier GSDB Code:	A1234	Design Lead:	Ford		
Part / System Name:	Ring - Oil Control	Ford Customer Plant:	Saarbrücken Assembly Plant - Germany		
CSPC:	050505	DFMEA Reference:	2021 B123 - Oil Control		
Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA: Global FMA Website (Ford Internal Only)		From DFMEA: YC Sev 9.10 : Potential CC. YS Sev 5-8 and may require special control to maintain process capability: Potential SC.	From PFMEA: CC Sev 9.10 (Part) : Critical Characteristic. SC Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.	OS Sev 9.10 (Process) : Operator Safety. HI Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.	
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS.		YC 4 YS 6	CC 0 SC 0	OS 0 HI 0	
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>.					
Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at each new data release.					
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method
1	Sheet Steel Material Properties	WSS-MLA365-A21	YC		
2	Sheet Steel Material Thickness	1.8 mm Thickness to ESBAU5-1N260-A	YC		
3	Motor Crank Nut Torque	25Nm ± 2Nm	YC		
4	Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS		
5	Mounting hole position	X345 Y150 Z-68 Positional tolerance Ø1.6mm	YS		
6	Bush Push In Force	8kN to 12kN	YS		
7	Wire Diameter	Ø 12.7mm ± 0.2mm	YS		
8	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	YS		
9	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS		
10	Material Properties (Flammability - FMVSS 302)	Carpet Surface: WSS-MBP16-B. Foam Underlay: WSS-M99P32-D1 Insulator: WSS-M99P32-D1	YC		

2. SCCAF Assessment V1 tab (External Only) completed and shows 100% Complete.

V1 External SCCAF Assessment for 2018 B123 MCA - Ring - Oil Control - John Doe Inc. - Pleasantville Plant - USA			
Current Status:	PASS	Milestone Report Status:	COMPLETE
Assessor	Graham Monks	Date	7-Oct-2018
Notes on Assessment	Assessment Completion Completion % 100% 20 Questions Answered out of 20 total. 0 Questions remain to be answered.		
Exit Full Screen			

3. All required "SCCAF Objects" have been created using "Create SCCAF" functionality in "SCCAFs" screen.
4. SCCAF with completed LHS (Characteristic, Spec & Tolerance, DFMEA class), , has been stored in the "Edit SCCAF" screen of its specific "SCCAF Object" in FEDE, do not store to "Attachments" screen. Note that the Header is auto populated from "Overview" screen in FEDE and cannot be updated from excel template.
5. **External SCCAF:**
 - a. SCCAF Assessment (V1 Tab) for each External SCCAF has been completed (100%) and loaded to "SCCAF Assessment" field on "Attachments" screen in FEDE. SCCAF cannot be published without SCCAF Assessment stored correctly and 100%.
 - b. All SCCAFs have been submitted for approval in the "Request Approval" screen in FEDE this will create a "Published" & locked version of the V1 / Unit PTC SCCAF Documents. No actual approval required @ V1.
 - c. SCCAF has been cascaded to the Suppliers (External).
 - d. SCCAF has been cascaded to the relevant STA Site Engineer (if required).
 - e. Number of SCCAFs completed will be reported automatically on the FMA Plan Item "SCCAFs" screen.
5. **Internal SCCAF:**
 - a. D&R Engineer submits their SCCAF to Workflow for transfer to Manufacturing Engineering.
 - o For Europe, IMG & China the Manufacturing Quality Engineer is included to review the SCCAF. Manufacturing engineering now have the write access to the SCCAF.

For help see [SCCAF Completion & Approval](#) & for Internal see [FEDE Help File - SCCAF Enhanced Workflow Process](#).

- YC can only ever become CC.
- YS can only ever be confirmed as SC or Blank.
- OS and HI do not have DFMEA class.
- No other classifications conversions are allowed.
- YS only becomes an SC or Blank through the PFMEA Occurrence rating.
- Confirmed as an SC if PFMEA rating is >3 or Blank if ≤3.
- SC can come from Ford Manufacturing / Supplier but **Must** have blank DFMEA class.

DFMEA Classification	PFMEA Occurrence rating	PFMEA Classification
YS	> 3	SC
YS	≤3	Blank

Suggested PFMEA Occurrence Evaluation Criteria		
Likelihood of Failure	Criteria: Occurrence of Cause – PFMEA (Incidents per items/vehicles)	Rank
Very High	≥ 100 per thousand	10
	≥ 1 in 10	
High	50 per thousand	9
	1 in 20	
	20 per thousand	8
	1 in 50	
	10 per thousand	7
	1 in 100	
Moderate	2 per thousand	6
	1 in 500	
	.5 per thousand	5
	1 in 2,000	
	.1 per thousand	4
	1 in 10,000	
Low	.01 per thousand	3
	1 in 100,000	
	≤.001 per thousand	2
	1 in 1,000,000	
Very Low	Failure is eliminated through preventive control.	1

Occurrence ratings can vary for different global manufacturing locations even for the same Supplier, or within different Ford Manufacturing locations even for the same processes. Each manufacturing location must complete a unique PFMEA.

The number of SCs for a similar component may vary for the different manufacturing locations and this will be documented on the specific SCCAF for that location.

- Ford Manufacturing or an existing Supplier that already has a similar component in production in the same location may have relevant production data to support an Occurrence rating ≤3 so a YS becomes Blank.
- A new manufacturer / new manufacturing location may not have this data so occurrence will be rated >3 and therefore a YS results in a PFMEA classification of SC.

If Occurrence rating is ≤3 LEA D&R Engineer should understand evidence for rating.

- For Internal SCCAF : If there is a new process or characteristic, then occurrence ≤3 is unlikely as there will be no data to support this rating.
- For External SCCAF : If there is a new manufacturing location or Supplier then occurrence ≤3 is unlikely, as the Supplier will not have the data to support their rating for that location. There may be new processes, machinery, operators or other factors that influence the occurrence with no historical data to support the rating.
- Quality History will also help in establishing previous issues to establish occurrence but must be from the same or very similar process and the same Supplier location or Ford Manufacturing location.

Evidence should comprise data that demonstrates acceptable process capability and comes from the same Product Characteristic:

- On a similar part.
- In the same facility/manufacturing environment.
- Using similar equipment & using similar process.
- Using the same supply chain.
- Measured using a capable measurement system and have the appropriate resolution to measure; the sensitivity to detect changes in the product or process variation (10 to 1 is the rule of thumb as per MSA AIAG).

Removal of SC – Proven Process Capability

For External Processes : Once a process can be shown to be capable then, in consultation with the STA site engineer, the Supplier can request removal of an SC from the Control Plan through the SREA process after the black-out period (e.g. Job1 + 90 days).

For Internal Processes : Once a process can be shown to be capable and/or the occurrence has proven to be ≤ 3 , then Ford Manufacturing can request removal of an SC from the Control Plan through a WERS Concern after the black-out period (e.g. Job1 + 90 days).

The line item would not be removed from the SCCAF, as the SCCAF is not required to be updated. The change will be completed in the PFMEA and Process Control Plan.

The removal of the SC does not mean that the actual Process Control Method is removed but it removes the requirement to monitor and document and can be measured at a reduced frequency / sample size.

YS not confirmed as an SC in SCCAF

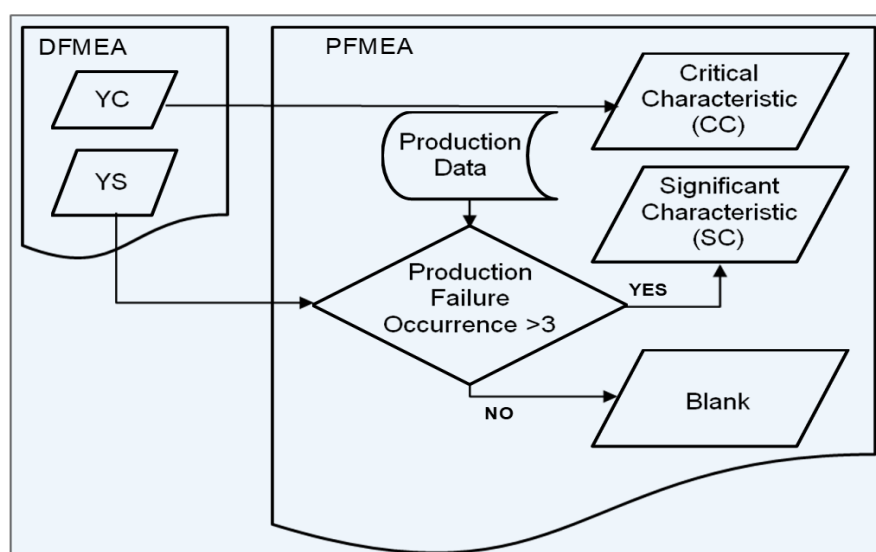
If YS is not confirmed as an SC, then select “-” in the PFMEA Class column and the existing Process Control Method must be entered into the Process Control Method column alongside either:

- The **actual** Ppk value for that process in that specific manufacturing location that the DFMEA to PFMEA conversion decision was based on (must be >1.33 ongoing production).
- The **actual** occurrence rating from the PFMEA for the relevant “potential cause” for that specific manufacturing location (must be less than or equal to 3).

Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method
Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS	-	CMM PPk=3.45
Wire Diameter	Ø 12.7mm ± 0.2mm	YS	-	Vernier Caliper PFMEA Occurrence = 3

If a YS is not confirmed as an SC, then the Product Characteristic should **not** be removed from the SCCAF, it is important for the following reasons:

- Traceability from DFMEA to signed SCCAF.
- To ensure that follow-on and future programs maintain and analyse all Product Characteristics.
- Each individual manufacturing location will review occurrence in the PFMEA and therefore whether a YS is confirmed as an SC.



Correct Classification Conversion

It is important that severity is checked for a YC and YS Product Characteristic and ensure correct confirmation of an SC or CC from the manufacturing process.

Teams should ensure that Ford Manufacturing / Suppliers complete the analysis for each Product Characteristic and review the PFMEA occurrence with associated data.

Incorrect ratings can lead to:

- YC being converted to an SC, which should not happen.
- YS not being confirmed as an SC when they should be.
- YS becoming OS or HI, which should not happen.
- YS being confirmed as an SC that should not be, with a potential for additional unnecessary cost.

Where a process can be shown to be stable and capable whilst using existing controls then the process does not need to be designated as an SC. SC controls attract extra cost associated with additional documentation and demonstration requirements through PPAP and ongoing production (see [p46](#) for more information).

Where an existing process can be shown to be stable and capable then such cost/work is not value added, so these characteristics should not be converted to SCs. There are two primary cases where teams should carefully consider YS to SC conversion to avoid unnecessary SCs:

1. Where the supplier currently produces a similar component at the same manufacturing location, and the intended process for the new part is common and proven on the existing process (with data); then it is likely that some or many YS characteristics will not convert to SCs as existing controls will be shown to be adequate. The team should carefully review all the YS to SC conversions in the SCCAF for the new part to understand the reason why the SC has been confirmed and if data from the existing process is applicable.
2. Where a supplier currently manufactures the part and the SCCAF is being updated for only minor changes, it is likely that some of the previously agreed SC characteristics may no longer need to be SCs as existing controls will be able to be shown to be adequate with data that did not exist when the original SCCAF was agreed. The team should review all YS to SC conversion in the SCCAF including those not subject to the change as a part of agreeing the updated SCCAF to ensure that the minimum set of SCs necessary with the data currently available is agreed.

Important Note: The PFMEA is the governing document that confirms the SC classification through the review of the historical occurrence of all causes associated with a failure mode. PFMEAs must be reviewed and updated using the latest data from the process. The PFMEA informs and confirms SC classifications in the SCCAF. SCs can ONLY be generated from the PFMEA.

FAA03-111-B - Special Characteristics Classification Table

[\(Link\)](#)

FAA-03-111-B : DFMEA / PFMEA Special Characteristics Table									
	Design FMEA				Process FMEA				
	Classification	To Indicate	Criteria	Actions Required	Classification	To Indicate	Criteria	Actions Required	Key Points
Customer (External) Effect	YC	Potential Critical Characteristic	Characteristic has a causal relationship to Potential Failure Modes having Severity of Effects rated 9-10	The Design and Release Engineer and the Manufacturing Engineer, and/or Supplier work collaboratively to develop optimal counter-measures	CC or V	Critical Characteristic	Characteristic aligns to Potential Failure Modes with Severity of Effects ratings of 9 & 10	Special Controls and Inclusion in the Control Plan See Section 7.5 Determination of a Suitable Process Control Strategy	- All CC's must be aligned to YC's in the DFMEA - All YC's must result in at least one CC somewhere in the process. Ideally, the CC should be identified at a point in the process where it is most effectively controlled.
	YS	Potential Significant Characteristic	Both criterion #1 and #2 must be met: 1. Characteristic has a causal relationship to Potential Failure Modes having Severity of Effects rated 5-8, or where agreed by the cross-functional team, having Severity of Effects rated <5 2. Characteristic may be influenced by the manufacturing process and may require special control to maintain the required process capability	The Design and Release Engineer and the Manufacturing Engineer, and/or Supplier work collaboratively to develop optimal counter-measures	SC	Significant Characteristic	Either criterion #1 or #2 are met: 1. Characteristics that aligns to Potential Failure Modes with Severity of Effects ratings of 5 to 8, with final PFMEA Occurrence Ratings greater than 3 2. Where the cross-functional team approves, characteristics that align to Potential Failure Modes with Severity of Effects ratings < 5, with final PFMEA Occurrence Ratings greater than 3	Special Controls and Inclusion in the Control Plan See Section 7.5 Determination of a Suitable Process Control Strategy	- The determination of whether or not YS's become SC's is based on estimation of rate of Occurrence using documented process capability data - SC's may be identified in the PFMEA, without alignment to a YS in the DFMEA, as long as they satisfy the PFMEA SC Criteria
					Blank	Not a Special Characteristic	Either criterion #1 or #2 are met: 1. Characteristics that aligns to Potential Failure Modes with Severity of Effects ratings of 5 to 8, with final PFMEA Occurrence Ratings less than or equal to 3 2. Where the cross-functional team approves, characteristics that align to Potential Failure Modes with Severity of Effects ratings < 5, with final PFMEA Occurrence Ratings less than or equal to 3		
Manufacturing / Assembly (Internal) Effect					OS	Operator Safety characteristic	Characteristic aligns to Potential Failure Modes with Severity of Effects ratings of 9 & 10	Safety Sign-off	
	Blank	Not a Potential Special Characteristic			HI	High Impact Characteristic	Characteristic aligns to Potential Failure Modes with Severity of Effects ratings of 5 to 8, with final PFMEA Occurrence Ratings greater than 3	Special Controls and Inclusion in the Control Plan See Section 7.5 Determination of a Suitable Process Control Strategy	- OS and HI characteristics MUST NOT align to characteristics designated in the DFMEA as YC or YS

Critical Characteristics are designated with the inverted delta symbol (∇) and are those product requirements (dimensions, functional performance requirements, material specifications, etc.) or process parameters (rates, temperatures, pressures, etc.) that can affect compliance with government regulations, or safe vehicle, or product function.

Critical Characteristics require specific manufacturing, assembly, shipping and/or monitoring action and inclusion in the Control Plan.

Controls for CC Characteristics

Overarching principle: CCs require Special Controls to prevent shipment of non-conforming product at all times.

These Special Controls are managed by the responsible tier, and are documented on the SCCAF, identifying the tier level of the control regardless of the supply chain tier where the CC is manufactured.

The following list (1-5) are the recommended Process Control Methods in order of preference, the top method being the most acceptable.

A: Controls for CC Characteristics at Point of Manufacture/Assembly

The controls are incorporated close to the point of manufacture of the CC, regardless of supply chain tier of manufacture.

A1: Poka Yoke Controls at Point of Manufacture

A method of ensuring that the manufacture or assembly can only be completed correctly or makes defects easily identifiable so they are not passed down to the next operation and never reach the customer. Poka-Yoke Controls can be implemented as elements of the manufacturing process as well as features incorporated into the product design.

A2: Manufacturing Process Control Approach

Where it has been consistently demonstrated that control of manufacturing process variables (KPIVs) can be 100% effective in controlling Critical Characteristics, CC Special Controls may focus on these manufacturing process variables. Some examples are: Material Specifications; Weld parameter control; Heat Treat Furnace controls; and Plating process parameters (see p35 for more details on KPIVs). Manufacturing process parameter control is used, with the product CC requirement validated at a frequency based on rational sampling that prevents shipment of non-conforming product.

Batch & Hold: Is a process control method that requires a batch of parts to be put on hold until appropriate sampling can be completed to confirm part quality and prevent the shipment of non-conforming product. There are typically two forms of batch and hold processes:

- **Batch sampling with a batch and hold process.**

This process is typically used for processes where several parts are processed at the same time (in a batch, sometimes referred to as a load) with the same piece of equipment (e.g. gear vacuum carburizing). Strategic sampling points within the batch/load based on the position within the load are defined to validate the quality of the entire load. If the part quality checks for the sampling points are in specification, the batch is released, otherwise the defined reaction plan for the affected process and characteristic(s) must be followed.

- **First and last sampling with a batch and hold process.**

This process is typically used for production processes such as flexible machining lines or transfer lines, where parts are produced in series. In this situation, a batch may be defined by a time window or piece count but may be reduced in size based upon an identified event (such as a tool change or machine crash). The first piece and last piece of the batch are checked to validate part quality. If the first and last part quality checks are in specification, the batch is released, otherwise the defined reaction plan for the affected process and characteristic(s) must be followed.

Additionally, the end item CC must remain as a CC to ensure appropriate reaction plans are in place.

Note: the bulleted methods above cannot be used independently of the process controls for a CC.

A3: Inspection/Measurement at Point of Manufacture

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Critical Characteristics are 100% measured at the relevant operation/station or immediately after manufacture/assembly and defective parts are clearly identified and scrapped or re-worked / reprocessed through the approved process to meet specifications. Any reworked products must repeat and meet all CC checks.

B: Controls for CC Characteristics After the Point of Manufacture/Assembly

B1: Inspection/Measurement at End of Manufacturing/Assembly Process

Critical Characteristics are 100% measured at end of line or equivalent and non-conforming parts are clearly identified and scrapped or re-worked / reprocessed through the approved process to meet specification. Any reworked products must repeat and meet all CC checks.

B2: Inspection/Measurement (by Ford Manufacturing / Supplier) Prior to Use

Critical Characteristics are 100% measured prior to use and non-conforming parts are clearly identified and scrapped or returned (e.g. incoming inspection). Any reworked products must repeat and meet all CC checks. Visual inspection (operator dependent) is not an acceptable Process Control Method.

Other Considerations for CC Controls

C: Destructive Testing

Where CC Special Controls are required for destructive testing, batch certification is used, including all three steps below:

- CC controls on Key Process Input Variables for all parts manufactured / assembled.
- Appropriate batch sampling for the destructive testing, to be agreed by Ford Manufacturing (internally supplied parts) or with STA site engineer and Supplier (externally supplied parts) or identified in Engineering Specs.
- Batch (all components manufactured between the first and last piece inspections) is held at Supplier / Internal Manufacturing location until positive test results are received.

D: Black Box / Grey Box (Supplier Only)

Where the CC is a black box / grey box performance requirement, two possible acceptable controls are:

- Where containable at the rate of usage of the black box components, validate the CC performance requirement for 100% of the black box components prior to the point of usage, or
- The Tier 1 Supplier ensures close management and monitoring of the Special Controls for the black box at the point of manufacture of the Product Characteristics, regardless of the tier level of manufacture, thereby ensuring compliance of the CC performance requirement. These Special Controls typically include poka-yoke at the point of manufacture of the CC, regardless of the tier level of manufacture, but may also use a Manufacturing Process Control Approach through control of Key Process Input Variables of all the parts manufactured / assembled.

E: Materials Designated as Critical Characteristics

Where a material has been designated as a CC, the following apply:

The effective control for material as a CC is the control of the usage of the correct material, typically including:

- Material certification by the bulk material Supplier to the designated Ford material specification.
- Poka-Yoke identification of the material to ensure the correct material is used (examples of poka-yokes for this purpose may include: bar coding with vision system, colour of material, hole pattern on blanks, etc.).
- Material handling processes to ensure use of only the correct material throughout manufacturing and assembly.
- Robust manufacturing / assembly process design to prevent usage of the incorrect material (e.g. stamping tooling designed such that blanks of a specific size are required to stamp a part and are only stocked in the correct material, so checking incoming blanks would provide sufficient control).

Note: this assumes that as a part of the design and verification process that the system/component performance has been verified to meet performance requirements for the full range of the material specification.

Significant Characteristics are those product, process, and test requirements that are important for customer satisfaction. Significant Characteristics require Quality Planning actions that must be addressed in a Control Plan.

SC items wherever possible should always be measured variably and charted using SPC as this will provide the best opportunity to assess if process performance is predictable and capable to meet customer expectations. For External SCCAFs visual inspection (operator dependent) is not an acceptable Process Control Method. For VME SCCAFs in very limited cases visual inspection may be accepted by agreement with responsible PD Engineering and VME team.

As stated previously in section: **DFMEA to PFMEA Classification Confirmation** on (p27) of this reference guide, an SC can be removed once capability has been established after the black-out period post launch.

To be acceptable a process must be in a state of statistical control and capable to be better than the tolerance with acceptable capability. This will lead to economically sound decisions i.e. the removal of unnecessary measurements.

Special Controls for SC and HI Characteristics

Special Control requirements for Significant Characteristics (SC) and High Impact (HI) Characteristics are exactly the same.

An appropriate Process Control Method is determined based upon the type of Special characteristic (discrete vs continuous), however, Poka Yoke controls (p30) and/or error proofing devices that prevent and/or detect non-conformances and prevent them from being passed on are always the preferred method when possible:

A: Discrete (Attribute) Characteristics:

Examples of Product Characteristics that are of the discrete type are: Presence; Absence; Orientation; Material Type, etc.

- Where poke yoke/error proofing is not feasible, other possible methods that detect non-conformance may be used, such as:
 - In-station control.
 - 100% automatic inspection.
 - Automated visual inspection and control (vision system).
 - 100% Key Process Input Variable (KPIV) Process Control Method with high frequency rational sample attribute measurements or Batch & Hold (review with responsible site STA / VME core / Plant Quality Manager for agreement).
- Gauging method will typically be of the attribute type.
 - Inclusion of guard banding with the attribute measurements. Refer to Ford PPAP Customer Specifics for more details on attribute gauge qualification.

Note: Guard banding is a reduction in gauge acceptance tolerance, tighter than the engineering specification, based on the measurement system error.

Note: Effective control method selection must consider process variability.

B: Continuous (Variable) Characteristics:

Examples of Product Characteristics that are of the continuous type are: Geometric Dimensions; Product Functional Parameters; Process Parameters, etc.

- Most continuous characteristics vary over time and should be monitored and controlled using Statistical Process Control (SPC) to achieve a state of statistical stability and control and allow for effective and timely identification of out of control conditions.
- Sampling frequency shall be based upon process capability and a rational sampling strategy. See the AIAG SPC manual for more information.
- Gauging method will typically be of the variable type.
- If attribute gauging is used, guard banding should be considered.
- For continuous characteristics related to process parameters, automated control is acceptable, coupled with rational sample product measurements, where proven to be effective.

Operator Safety Characteristics (OS) are related to Process Parameters or Product Characteristics that may adversely affect the safety of the operator or compliance with Governmental Regulations (e.g., Occupational Safety and Health Administration [OSHA] requirements, Ford Health and Safety Specifications, etc.). These Special Characteristics are required to be included in a safety sign-off as designated by local regulation. The safety sign-off may be completed through the safety risk assessment process for Internal Ford processes.

High Impact Characteristics (HI) are related to process parameters or product characteristics that can adversely affect the operation of the process or subsequent operations, but that do not adversely impact customer satisfaction.

HI Characteristics are indicated by process failure modes with failure mode severity from 5 to 8 and occurrence greater than 3.

High Impact Characteristics require emphasis in Quality Planning actions that must be listed in a Control Plan.

OS and HI Characteristics relate to the process only, they exist only in the PFMEA – there is no potential OS or potential HI in the DFMEA, therefore they **must** have Blank in DFMEA Class Column on the SCCAF.

Example of HI Characteristic

A hole added to a sheet metal part to aid in the stamping process as a locator for multi-die stamping. This hole has no influence on the actual performance of the part but if not formed or in the correct position will affect the downstream manufacturing process. Therefore will require emphasis in the Quality Planning actions.

Required Information for Process Control Method

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The Process Controls listed must be the Production Controls (and frequencies if required) planned to be in place for production. Process Controls should not be used to PPAP and then removed for production.

CC:

Need to include actual Process Control Method and the frequency as this forms part of the Process Control Method, such as:

- 100% inspection.
- KPIV Process Parameter 100% control with First Part / Last Part measurement per batch with Batch and Hold.

Note: All CC controls for Internal Manufactured & External Suppliers torque related characteristics should always include torque **and** angle controls for each joint.

SC: Variable Measurement:

Need to include actual Process Control Method but **should not** include the frequency/sample size, as this will be established during process development through the unit or vehicle builds.

Frequency and sample size will be dependent on the variation of parts manufactured during the later build phases, selecting them at V2 / Unit TPVDJ potentially means that we are sampling too frequently for the variation (added cost), not regularly enough or not sampled at the right time to capture between subgroup variation (risk of out of spec components).

SC: Attribute Measurement:

Need to include actual Process Control Method (refer to point 6 on p34 for more information).

YS to Blank:

Include existing Process Control Method. Frequency is not required but include the actual PFMEA Occurrence rating or actual Ppk value to justify non-conversion of a YS.

1. **Supplier** (External) or **Ford Manufacturing** (Internal) must complete Process Control Method even if a Generic SCCAF has the Process Control Method column completed, the Generic stated method should only be taken as a recommendation.
2. Every line item requires an **ongoing production** Process Control Method. Any reference to DV / PV / EVP / Capability / PPAP/ Safe Launch etc. are not acceptable and should not be included (even if in addition to an adequate process control).
The SCCAF will help identify the characteristics to be measured on components through the early Build phases and should be the minimum measurements included into the Prototype & Pre-Production Control Plans and Measurement reports but DV/PV is not to be included on the SCCAF.
3. Process Control Method column should not reference changes to control methods or sample size / frequencies post Job 1, should only be the proposed method for start of production (Job 1). If Suppliers / Ford Internal Manufacturing want to change control methods after the blackout period (see [p28](#) for more details).
4. Any reference to tooling maintenance or reaction plans should not be included on the SCCAF, the Process Control Method column details how the part is being controlled/measured during production. Additional information such as tool changes, maintenance and reaction plans should be in the PFMEA & Process Control Plans. That detail is not required in the SCCAF.
5. All YS that convert to blank must have the actual Ppk value or actual PFMEA occurrence detailed in the Process Control Method Column for that Product Characteristic with the Process Control Method and that PFMEA occurrence should be able to be demonstrated with data (see [p28](#)).
6. All SC with attribute measurements (e.g. Go/No-Go gauge) should have a 100% inspection control method with guard banding (as required, see PPAP customer specific requirements). As capability of 1.33 (production) / 1.67 (pre-production) cannot typically be demonstrated for attribute data. However, a 100% Key Process Input Variable (KPIV) Process Control Method with high frequency rational sample attribute measurements or Batch and Hold can also be acceptable (review with responsible site STA / VME core / Plant Quality Manager for agreement).
There are very few Product Characteristics that are truly attribute. At a minimum, SPC for KPIV are required with the Batch and Hold process to be able to monitor any trends in production. Effective control charts for KPIV can be constructed even with small amounts of data. Alternatively, KPIV can be controlled 100% with an automated process.
7. All SC with variable measurements should either have SPC or Batch/Hold with Process Controls and is detailed in the Process Control Method Column for that Product Characteristic.
8. YS characteristics which utilise GD&T features (geometric characteristics) to express specification and tolerances for a characteristic are often not independent measurements, and as such require a detailed discussion with Supplier / Internal Manufacturing to understand the manufacturing process and establish appropriate measurement methods which includes trend analysis (SPC). Examples of dependent measurements are profiles, positional tolerances, perpendicularity and angularity etc. CMM should be used as the measurement tool for such characteristics as it can measure displacements.
9. Each Characteristic will need the actual detailed Process Control Method column completed and cannot just refer to other line items e.g. See item 3.
10. Process Control Method column should not include optional methods that the Supplier / Ford Internal Manufacturing might use e.g. "CMM or Vernier". All methods detailed should be planned to be used in production.
The SCCAF is used to agree the proposed control method that will be used in production, any changes to the proposed method should go through the process outlined in SCCAF update tables ([p8](#) & [p9](#)).

The Tier 1 Supplier is responsible for all levels of manufacture of their component or assembly and the process method to control the Product Characteristic must be included on the SCCAF regardless of Tier level e.g. Tier 2 / Tier 3 etc.

Statements such as “Controlled at Tier 2” or “Measured at Sub-Supplier” are NOT acceptable. The detailed Process Control Method must be listed just as required for a Tier 1. This approach ensures they have detailed the correct controls that will be appropriate for that Product Characteristic.

It is not intended for the CC/SC level controls to be repeated at each Tier of the manufacturing processes. If the Tier 2, 3 etc. applies the correct level of control mandated for a CC or SC then subsequent controls are not required at the Tier 1.

The LEA D&R Engineer may request to review the lower Tier level documents such as PFMEA and Process Control Plans. Note that it is mandatory for the D&R and STA engineer to sign off on CC Product Characteristics in the Control Plan and PFMEA (refer to Ford PPAP customer specific requirements for more information).

Characteristic : “Material Properties”

- Process Control Method would be “Material Certification per batch with material usage controls at manufacturing stations” at Tier 2, this is the appropriate level of control, this should be detailed on the SCCAF.
- We do not need to go further down the Supplier process, i.e. raw material.

Characteristic : “Hole diameter”

- Dimension and required Control would be measured at point of manufacture (Tier2) so the measurement method would be “Vernier, CMM, 100% Plug Gauge etc.”
- The Tier 1 will not have to re-verify the measurement or include their own Process Control Method.

In certain cases, the Tier 1 may also have a Process Control Method in place to verify the Product Characteristic, this should then be listed to fully understand the complete Process Control Method for the Product Characteristic identified.

The team can then identify the Tier 1 and Tier 2 / Tier 3 etc. specific controls in the Process Control Method column, some examples shown below.

№	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method
1	Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS	SC	Tier 2: 100% GO / NO GO Gauge Tier 1: Incoming Inspection 5 pcs / batch Internal Micrometer
2	Motor Crank Nut Torque	25Nm ± 2Nm	YC	CC	Tier 2: 100% torque control DC nut runner with interlock Tier 1: Incoming Inspection : 10pcs / batch Torque wrench
3	Glass Thickness	3.5mm ± 0.2mm	YC	CC	Tier 1: Lot Certification Glass thickness measured 1 part per load. Batch and hold and material tracking through the facility Tier 2: Float glass thickness measured 1pc per load Quality check measures formed glass thickness every two hours Tier 3: 100% process controlled by laser thickness check of float glass every 1.5 minutes across the width of the glass.
4	Blackout distance to glass edge (sub-component)	1.5mm maximum	YS	-	Tier 1: Lot traceability through production recording sheet Tier 2: Lot certification Attribute Scale. PFMEA occurrence: 3

Where direct measurement of Product Characteristics is infeasible or impractical, these Product Characteristics may be indirectly controlled through the control of relevant Key Process Input Variables (KPIV - Process Characteristics or Parameters).

Where it has been consistently demonstrated that control of manufacturing process variables can be 100% effective in controlling Critical Characteristics, CC Special Controls may focus on these manufacturing process variables.

Where Manufacturing process parameter control is used, the product CC requirement (Product Characteristic) must still be validated at a frequency based on rational sampling that prevents shipment of non-conforming product and included in the Control Plan.

Control emphasis is on prevention through control of Process Parameters. The Process Parameters must be included in the Process Control Method column of the SCCAF as well as the Control Plan, examples include:

- Control of a welding process through control of weld current, wire feed rate and weld path.
 - Confirm through a weld section using rational sampling.
- Control of electroplating through bath chemical composition, temperature, current and plating time.
 - Confirm through a corrosion test using rational sampling.

Process Characteristics are usually only added to the SCCAF by the manufacturing team with specification and tolerance identified and the DFMEA Classification column left blank.

If the PFMEA identifies a missed Design Product Characteristic, this should be added to the DFMEA and the DFMEA Classification column should be completed.

DFMEA				PFMEA	
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method
1	Welding Specification	Spec WSBCPA-1B310-AA	YC	CC	Full-Automatic Welding - Controlled by KPIVs Cut & Etch Test First Off - Last Off. 1 Cut section for every 50mm with batch & hold per stream / fixture of production
1a	Weld Wire Diameter - KPIV	$\varnothing 8 \pm 0.3\text{mm}$		CC	Process startup verification - Material data Sheet Sign Off Diameter Measurement - Vernier Caliper
1b	Weld Current - KPIV	$\pm 5\%$ nominal setting		CC	Current [A] is an automated measurement and is measured 100% by welding facility. Process startup verification
1c	Weld Wire Feed Speed - KPIV	$\pm 10\%$ nominal setting		CC	Weld Wire Speed [m/min] is an automated measurement and is measured 100% by welding facility Process startup verification
1d	Weld Travel Speed - KPIV	$\pm 10\%$ nominal setting		CC	Welding Speed [m/min] is an automated measurement and is measured 100% by welding facility Process startup verification

SCCAF Assessment – External Only

[Contents Page](#)

A tool to support the completion of SCCAFs with quality and to GPDS gateway timing. A collection of simple questions to evaluate the information contained in the SCCAF (Template & Training on [Global FMA Website - SCCAF](#)).

The questions are based upon lessons learnt from SCCAF reviews across all Functions and Regions.

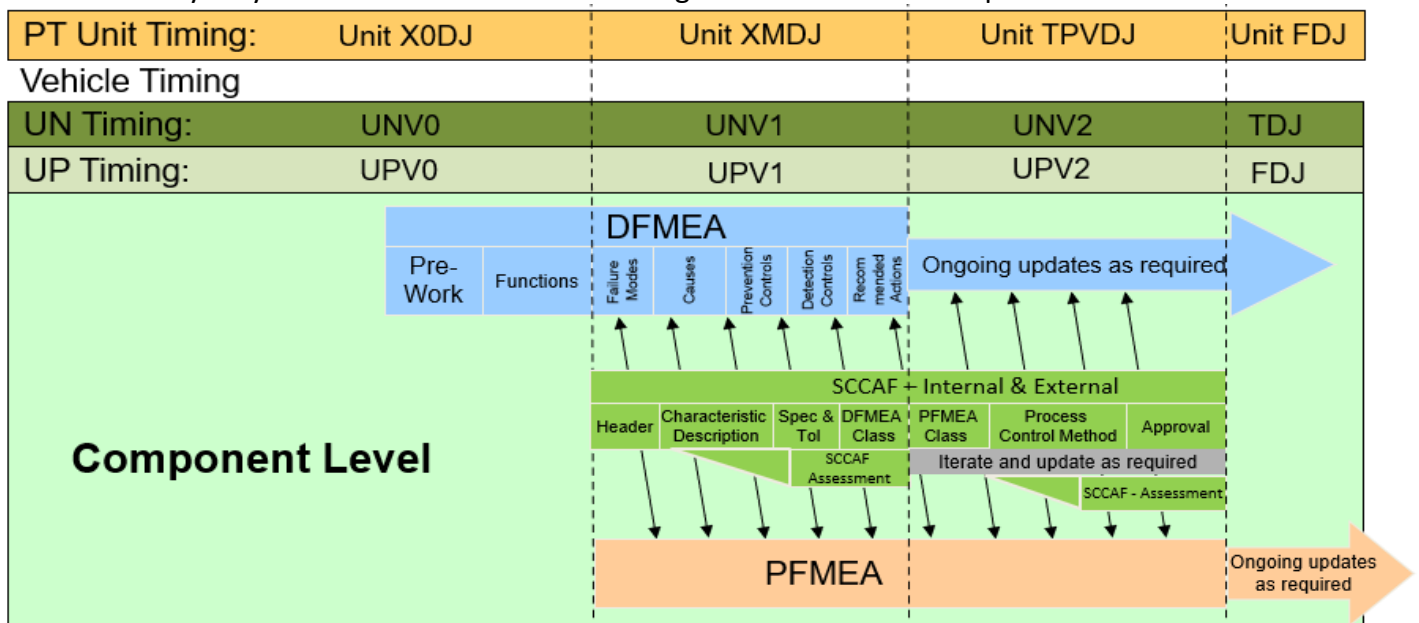
The SCCAF Assessment is owned and completed by LEA D&R Engineer in 2 stages:

- V1 / Unit PTC – Questions and Review of LHS SCCAF.
- V2 / Unit TPVDJ – Questions and Review of RHS SCCAF.

Note: Where the SCCAF LHS is common across multiple SCCAFs only one SCCAF Assessment will be required for V1 / Unit PTC deliverable completion but the analysis and review must be completed for all. Some examples include:

- Common design but multiple Suppliers.
- Multiple parts with common Product Characteristics e.g. Sheet metal Stamping.

However, for V2 / Unit TPVDJ a SCCAF Assessment will be required for all SCCAFs as Process Control Methods may vary and need to be reviewed and agreed between each required SCCAF.



V1 / Unit PTC Assessment

V1 External SCCAF Assessment for 2018 B123 MCA - Ring - Oil Control - John Doe Inc. - Pleasantville Plant - USA				
Current Status:	PASS	Milestone Report Status:	COMPLETE	Assessment Completion
Assessor:	Graham Monks	Date:	7-Oct-2018	Completion %: 100%
Notes on Assessment:	20 Questions Answered out of 20 total. 0 Questions remain to be answered.			
Exit Full Screen				
Question	Answer	Comments	Improvement Information - What and Why	
1 Is the External SCCAF form FAF03-111-2 used?	Yes			
2 Is only 1 Supplier location specified on form and includes the City and Country of the manufacturing location?	Yes			

V2 / Unit TPVDJ Assessment

V2 External SCCAF Assessment for 2018 B123 MCA - Ring - Oil Control - John Doe Inc. - Pleasantville Plant - USA				
Current Status:	PASS	Milestone Report Status:	COMPLETE	Assessment Completion
Assessor:	GRAHAM MONKS	Date:	8-Oct-2018	Completion %: 100%
Notes on Assessment:	17 Questions Answered out of 17 total. 0 Questions remain to be answered.			
Go Full Screen				
Question	Answer	Comments	Improvement Information - What and Why	
21 How many YCs are converted to CCs	1			
22 How many YCs are converted to Blanks	0			

For the V1 / Unit PTC & V2 / Unit TPVDJ gateways the engineering teams will answer each question. If not answered adequately then help information will be shown on RHS of the document to guide updates of the SCCAF to the required quality.

- “What you need to do to rectify the issue.”
- “Why it’s needed to be modified.”

The SCCAF will then need to be modified as per the help information and the D&R engineer must re-assess the question.

Section 1: LHS Characteristics : Specification & Tolerance : DFMEA Class DUE V1				
1	Is only 1 Supplier location specified on form and includes the City and Country of the manufacturing location?	No		Create a separate SCCAF for each manufacturing location Why: There must be a SCCAF for each unique manufacturing location as controls may be different for these different locations. If a single SCCAF was used for multiple locations, there is a risk that at some of the locations the controls may not be implemented or to the required level.
2	Is John Doe Inc. - Pleasantville Plant - USA the location where the part will actually be produced?	No		Modify Supplier location on the header Why: The SCCAF must be specific to the supplier manufacturing location that will actually produce the part and signed off by the management at that location. Why: the process controls need to be completed specifically to the manufacturing process, even nominally identical plants are often different in their controls or application.
3	Does this SCCAF cover multiple parts? For multiple parts: Are the parts substantially the same in design with only small differences?	Yes No		Create different SCCAFs for different parts. Multiple worksheets within the one workbook may be used. Why: SCCAFs need to ensure clear agreement on characteristics and controls for each part. Grouping parts with significant difference risks confusion. **NOTE: An exception to this may be available - discuss with PD Quality and use the columns to the right.**
4	Is part number and part name information completed (prefix, base and first letter of suffix as a minimum) for all parts covered?	No		Complete the part information Why: SCCAF does not clearly specify the part or parts covered. Risk that what is covered by the SCCAF could be interpreted differently leading to incorrect controls or no controls.

Exception Process- SCCAF Assessment

For some of the questions when the selected answer is RED the engineer has the option of requesting an exception. In these cases, this is noted in the “Improvement Required” column.

This exception request needs to be discussed and approved with Quality / Design Assurance team and the reasons clearly defined for the request.

				Exception Process	
Question	Hide Header	Go Full Screen	Answer	Improvement Information - What and Why	Exception Possible
4	Does this SCCAF cover multiple parts?		Yes		
	For multiple parts: Are the parts substantially the same in design with only small differences?		No	Create different SCCAFs for different parts. Multiple worksheets within the one workbook may be used. Why: SCCAFs need to ensure clear agreement on characteristics and controls for each part. Grouping parts with significant difference risks confusion. **NOTE: An exception to this may be available - discuss with PD Quality and use the columns to the right.**	If this issue cannot be resolved, an exception may be possible - please discuss with your PD Quality team to agree if an exception is acceptable.

COMPLETE: All questions completed with required quality.

Current Status:	PASS	Milestone Report Status:	COMPLETE	Assessment Completion
Assessor:	GMONKS	Date:	28-Oct-2017	Completion %
Notes on Assessment				100%
Exit Full Screen				33 Questions Answered out of 33 total. 0 Questions remain to be answered.

GREEN: SCCAF is on track to meet “Complete” status, continue to answer remaining questions.

Current Status:	PASS	Milestone Report Status:	GREEN	Assessment Completion
Assessor:	GMONKS	Date:	28-Oct-2017	Completion %
Notes on Assessment				59%
Exit Full Screen				19 Questions Answered out of 32 total. 13 Questions remain to be answered.

RED: Line item on SCCAF must be resolved and question re-assessed. Review the help information provided in the assessment.

Current Status:	FAIL	Milestone Report Status:	RED	Assessment Completion
Assessor:	GMONKS	Date:	28-Oct-2017	Completion %
Notes on Assessment				28%
Exit Full Screen				8 Questions Answered out of 29 total. 21 Questions remain to be answered.

Question	Answer	Comments	Improvement Information - What and Why
1	No		Create a separate SCCAF for each manufacturing location Why: There must be a SCCAF for each unique manufacturing location as controls may be different for these different locations. If a single SCCAF was used for multiple locations, there is a risk that at some of the locations the controls may not be implemented or to the required level.

The team needs to work closely with Ford Manufacturing Engineering (PTME, VME, or Plant). Only ongoing controllable Product Characteristics such as torques should be included.

DFMEA does not consider manufacturing errors, so process design issues such as correct handling should not be included as these are not Design Characteristics.

Manufacturing errors such as “Do not drop”, “No Pinch, Kink or Damage from Installation” should not be included in the SCCAF as they would not be identified in the DFMEA therefore not YS or YC. These should be reviewed through the manufacturing process documents such as PFMEA, Process Control Plans and work instructions. These items can be communicated using the optional “Manufacturing Information Transfer Sheet” ([p22](#)). Additional items maybe added on to the SCCAF that have been driven out of the PFMEA review. These items do not require a DFMEA classification as they are driven from the PFMEA and are process related and controls may necessitate the request of tools/gauging.

Internal SCCAFs may contain frequencies and sample size within the Process Control Method column for established agreed processes within the Internal Ford Manufacturing facility.

GSPAS / VPS Sheet #

New optional field added to support comparison between WERS / AVBOM / SCCAF & GSPAS supporting close loop with control plan. Format should align with Process Sheet Number: 3 Letters and 5 Numbers e.g. FHN51940.

PFMEA		
PFMEA Class	Process Control Method	GSPAS / VPS Sheet #

SCCAF Approval Methods

Approval using DocuSign

[DocuSign](#) is recommended for approval of SCCAFs as it will create a single document showing approval from all signatories; this must then be stored in FEDE as evidence and shows traceability of completion.

Help files on completing DocuSign for SCCAFs is available on [Global FMA Website - SCCAF](#).

- [DocuSign SPL - SCCAF Sign Off](#) - Help on how to complete the electronic sign off process for SCCAF documents using DocuSign.
- [DocuSign SPL - SCCAF DocuSign Templates](#) - Help on using pre-created SCCAF Templates within DocuSign that automatically generates the required fields selected.

Note:

- For **External** SCCAFs there are some Country restrictions for use of DocuSign outside of Ford (review [Ford DocuSign Info Webpage](#) for countries listed), for further details and recommended process for usage of DocuSign in these cases see [DocuSign SPL - SCCAF Sign Off](#).
- All **Internal** SCCAFs can be approved using DocuSign.

The LEA D&R Engineer is responsible for ensuring the approval of all manufacturing locations for both External and Internal SCCAFs.

FAF03-111-2 : Externally Supplied Content

Special Characteristics Approvals required at UNV2 / UPV2 / PA as per FAP 03-111				
Ford D & R Engineer Approval / Date:	Signature	Name	e-mail	Date
Ford STA Engineer Approval / Date:	Signature	Name	e-mail	Date
Ford Craftsmanship Approval / Date:	Signature	Name	e-mail	Date
Supplier Plant Quality Manager Approval / Date:	Signature	Name	e-mail	Date

Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocSign help & DocuSign Country Restrictions).

Ford D&R Engineer:

- **Mandatory** Approval by LEA D&R Engineer.

Ford STA Engineer:

- **Mandatory** Approval by Supplier Plant specific STA Site Engineer for:
 - PPAP Level 3 and 5 Suppliers.
 - or
 - All Priority Suppliers.
 - or
 - When requested by the LEA D&R Engineer regardless of the PPAP Submission Level.

Ford Craftsmanship:

- Is only required for parts subject to Craftsmanship / Appearance approval.

Supplier Plant Quality Manager:

- **Mandatory** Approval by Supplier Plant Quality Manager for Supplier Manufacturing location relevant to each specific SCCAF.

FAF03-111-1 : Internal Ford Manufactured and Assembled Content

Special Characteristics Approvals required at UNV2 / UPV2 / PA as per FAP 03-111				
Ford D & R Engineer Approval / Date:	Signature	Name	e-mail	Date
Ford Manufacturing Engineer Approval / Date:	Signature	Name	e-mail	Date
Ford Craftsmanship Approval / Date:	Signature	Name	e-mail	Date
Ford Plant Quality Manager Approval / Date:	Signature	Name	e-mail	Date

Note: Approvals may be obtained by physical signature or electronic sign-off. Recommended: Use DocuSign to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment (See SCCAF FMA webpage for DocSign help).

Ford D&R Engineer:

- **Mandatory** Approval by LEA D&R Engineer.

Ford Manufacturing Engineer:

- **Mandatory** Approval by local Ford Manufacturing Engineer for the specific manufacturing location relevant to SCCAF.

Ford Craftsmanship:

- Is only required for parts subject to Craftsmanship / Appearance approval.

Ford Plant Quality Manager:

- **Mandatory** Approval by local Ford Plant Quality Manager for Ford Manufacturing location relevant to each specific SCCAF.

For more information on FMA GPDS Deliverables including SCCAF : [FMA Deliverables Table - GPDS Gateways](#).
For more details on how to store SCCAFs in FEDE when using Global FMA Tracker : [SCCAF Storage](#) document on [Global FMA Website - SCCAF](#) or [FEDE Help SPLs](#).

1. PFMEA Class is completed and the defined Process Control meets the requirements for the relevant PFMEA classification (CC,SC etc.) as per Ford Guidelines.
2. SCCAF Assessment – V2 tab (External Only) completed and shows 100% Complete.
3. SCCAF has been signed off by all required parties ([DocuSign](#) is recommended, see info on [p40](#)).

Special Characteristics Communication and Agreement - External				OPTIONAL: SCCAF Draft Vers: Date:	
Lead / Support roles and timing for each stage	Ford PD engineer / Supplier	Supplier / Ford PD	SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section SCCAF Handbook SCCAF Global FMA Website		
Model Year & Program Code:	2021 B123 MCA	Part / System Number:	BC3Q-6153-A		
Supplier Name / Plant:	John Doe Inc. - Pleasantville Plant - USA	Engineering Release Number:			
Supplier GSDB Code:	A1234	Design Lead:	Ford		
Part / System Name:	Ring - Oil Control	Ford Customer Plant:	Saarlouis Assembly Plant - Germany		
CPSG:	050505	DFMEA Reference:	2021 B123 - Oil Control		
Special Characteristics for: 2021 B123 MCA John Doe Inc. - Pleasantville Plant - USA BC3Q-6153-A			(Suppliers Only - Accessible through Ford Supplier Portal). (Ford Internal Only)		
Ford D & R Engineer Approval / Date: <i>AN Other</i> Ford STA Engineer Approval / Date: <i>Ann Engineer</i> Ford Craftsmanship Approval / Date: <i>A Part</i> Supplier Plant Quality Manager Approval / Date: <i>A Part</i>			Approval Signature Name e-mail Address Approval Date <i>AN Other</i> AN Other another@ford.com 08-Mar-23 <i>Ann Engineer</i> Ann Engineer annengineer@ford.com 09-Mar-23 <i>A Part</i> A Part apart@johndoeinc.com 10-Mar-23		
Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).					
Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA: Global FMA Website (Ford Internal Only)					
From DFMEA: YC Sev 9,10 : Potential CC. YS Sev 5-8 and may require special control to maintain process capability: Potential SC.		From PFMEA: CC Sev 9,10 (Part) : Critical Characteristic. SC Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.		OS Sev 9,10 (Process) : Operator Safety. HI Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.	
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS.		YC 4 YS 6 CC 4 SC 4 - 2 OS 0 HI 0			
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>. Special Characteristics Communication and Agreement for all UP content is required by UPV2. Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.					
DFMEA			PFMEA		
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC	CC	Material Certification per batch with material usage controls at manufacturing stations
2	Sheet Steel Material Thickness	1.8 mm Thickness to ESBAU5-1N260-A	YC	CC	Material Certification per batch with material usage controls at manufacturing stations
3	Motor Crank Nut Torque	25Nm ± 2Nm	YC	CC	100% torque control DC nut runner with Interlock
4	Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS	-	CMM PPK=3.45
5	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	SC	100 % go / no go fixture
6	Bush Push In Force	8kN to 12kN	YS	SC	100% in process with load cell
7	Wire Diameter	Ø 12.7mm ± 0.2mm	YS	-	Vernier Caliper PFMEA Occurrence = 3
8	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	YS	SC	Paint Thickness Ultrasound Tester with SPC
9	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS	SC	100% EOL voltage tester
10	Material Properties (Flammability - FMVSS 302)	Carpet Surface : WSS-M8P16-B. Foam Underlay : WSS-M9P32-D1 Insulator : WSS-M9P32-D1	YC	CC	Material Certification per batch with material usage controls at manufacturing stations

4. SCCAF with completed RHS (Internal & External) has been stored in the “Edit SCCAF” screen of its specific “SCCAF Object” in FEDE, do not store to “Attachments” screen.
5. SCCAF Assessment for each External SCCAF has been loaded to “Attachments” screen in FEDE.
6. Evidence of approval has been loaded to “Approval Evidence” field on “Edit SCCAF” screen.
7. All SCCAFs have been submitted in the “Request Approval” screen in FEDE this will create a “Published” and locked version of the V2 / Unit TPVDJ SCCAF Documents.
8. SCCAFs have been marked as “Loaded to FEDE” and “Complete” in “Global FMA Tracker”.

Internal SCCAFs:						
Required	Cascaded to VOME @ V1	With PD for Update	With VOME	With Plant Mgr	Complete	Loaded to FEDE
1	1	0	0	0	1	1
External SCCAFs:						
Required	Cascaded to Supplier @ V1	With PD for Update	With Supplier	With STA	Complete	Loaded to FEDE
2	2	0	0	0	2	2

For help see [SCCAF Storage](#) document.

For more information on FMA GPDS Deliverables including SCCAF : [FMA Deliverables Table - GPDS Gateways](#).
For more details on how to store SCCAFs in FEDE when using **FMA Plan in FEDE** : [SCCAF Completion & Approval](#) document on [Global FMA Website - FMA Plan](#) or [FEDE Help SPLs](#).

1. PFMEA Class is completed and the defined Process Control meets the requirements for the relevant PFMEA classification (CC,SC etc.) as per Ford Guidelines.

Special Characteristics Communication and Agreement - External				OPTIONAL:	SCCAF Draft Vers:	Date:
Lead / Support roles and timing for each stage	Ford PD engineer / Supplier	Supplier / Ford PD	SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section SCCAF Handbook - SCCAF Global FMA Website	Special Characteristics for: 2021 B123 MCA John Doe Inc. - Pleasantville Plant - USA BC3Q-6153-A		
Model Year & Program Code:	2021 B123 MCA	Part / System Number:	BC3Q-6153-A	Ford D & R Engineer Approval / Date: <i>Alt Other</i> AN Other another@ford.com 08-Mar-23		
Supplier Name / Plant:	John Doe Inc. - Pleasantville Plant - USA	Engineering Release Number:		Ford STA Engineer Approval / Date: <i>Ann Engineer</i> Ann Engineer annengineer@ford.com 09-Mar-23		
Supplier GSDB Code:	A1234	Design Lead:	Ford	Ford Craftmanship Approval / Date: <i>A Part</i> A Part apart@ohndoeinc.com 10-Mar-23		
Part / System Name:	Ring - Oil Control	Ford Customer Plant:	Saarlouis Assembly Plant - Germany	Supplier Plant Quality Manager Approval / Date:		
CPSC:	050505	DFMEA Reference:	2021 B123 - Oil Control	Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).		
Key:	For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA: Global FMA Website (Ford Internal Only)	From DFMEA:	From PFMEA:	OS Sev9.10 (Process) : Operator Safety. HI Sev5-8 & Occ 4-10 (Process) : High Impact Characteristic.		
Special Characteristic Totals:	Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS.	YC 4 YS 6	CC 4 SC 2	OS 0 HI 0		
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>.						
Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.						
DFMEA			DFMEA Class	PFMEA Class	PFMEA	
No.	Characteristic Description	Specification & Tolerance			Process Control Method	
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC	CC	Material Certification per batch with material usage controls at manufacturing stations	
2	Sheet Steel Material Thickness	1.8 mm Thickness to ESBAU5-1N260-A	YC	CC	Material Certification per batch with material usage controls at manufacturing stations	
3	Motor Crank Nut Torque	25Nm ± 2Nm	YC	CC	100% torque control DC nut runner with Interlock	
4	Mounting hole diameter	2.5mm +0.0mm, -0.03mm	YS	-	CMM PPK=3.45	
5	Mounting hole position	X345 Y150 Z68 Positional tolerance Ø1.6mm	YS	SC	100 % go / no go fixture	
6	Bush Push In Force	8kN to 12kN	YS	SC	100% in process with load cell	
7	Wire Diameter	Ø 12.7mm ± 0.2mm	YS	-	Vernier Caliper PFMEA Occurrence = 3	
8	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max 35 µm	YS	SC	Paint Thickness Ultrasound Tester with SPC	
9	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS	SC	100% EOL voltage tester	
10	Material Properties (Flammability - FMVSS 302)	Carpet Surface: WSS-M8P16-B. Foam Underlay: WSS-M9P32-D1 Insulator: WSS-M9P32-D1	YC	CC	Material Certification per batch with material usage controls at manufacturing stations	

2. SCCAF with completed RHS (Internal or External) has been stored in the "Edit SCCAF" screen of its specific "SCCAF Object" in FEDE, do not store to "Attachments" screen.
3. **External SCCAF:**
 - a. SCCAF Assessment (V2 tab) for each External SCCAF has been completed (100%) and loaded to "SCCAF Assessment" field on "Attachments" screen in FEDE. SCCAF cannot be published without SCCAF Assessment stored correctly and 100%.
 - b. SCCAF has been signed off by all required parties (DocuSign is recommended, see info on p40).
 - c. Evidence of SCCAF Approval loaded to "Approval Evidence" field on "Edit SCCAF" screen.
 - d. All SCCAFs have been submitted in the "Request Approval" screen in FEDE this will create a "Published" and locked version of the V2/ Unit TPVDJ SCCAF Documents.
 - e. SCCAF completed will be reported on the FMA Plan Item.
6. **Internal SCCAF**
 - a. Manufacturing Engineer completes RHS of SCCAF and send to D&R Engineer for V2 approval.
 - b. D&R Engineer approves SCCAF through FEDE approval process.
 - c. Ford Craftmanship approves SCCAF through FEDE approval process (if required).
 - d. Plant Quality Manager approves SCCAF through FEDE approval process.
 - e. Number of SCCAFs completed will be reported automatically on the FMA Plan Item.

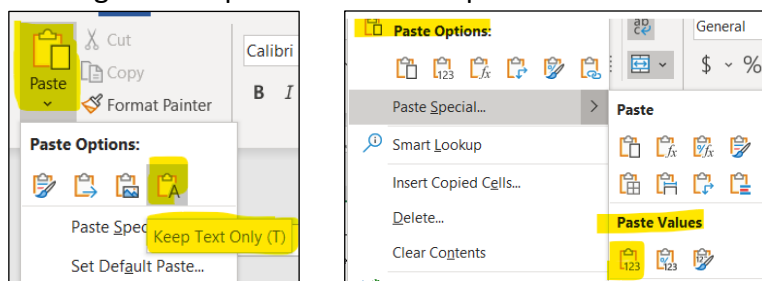
For help see [SCCAF Completion & Approval](#) document.

Many Core Engineering teams have a generic SCCAF that program application teams can use as a basis for the programs. These are in the process of being updated to the latest FAP03-111 template, all programs should be using the latest template shown in the SCCAF Handbook and available through either:

- [Global FMA Website - SCCAF](#)
- [FAP03-111 : Selection, Identification, and Control of Special Characteristics](#)

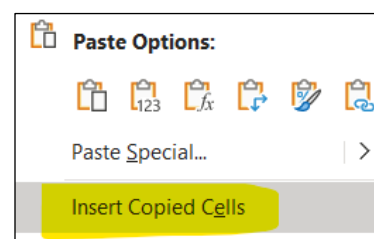
If teams need to transfer the information from a Generic SCCAF or a previous program they are using as a surrogate then below are some tips to ensure the formatting and checking functionality within the updated SCCAF template are maintained. The updated SCCAF templates has checks including but not limited DFMEA to PFMEA conversion & YC to CC counts.

- The SCCAF template is password protected and no additional columns should be added in the main SCCAF, this is a Global released FAP document and any modifications could risk a GAO Audit issue.
- Copy each section of the SCCAF one at a time e.g. “Characteristic Description” then “Specification and tolerance”.
- Most important is to use “Paste – Text Only” this will maintain the integrity of the conditional formatting on the updated SCCAF templates.



- To add additional rows then select a complete existing row then “Right mouse button” “Copy” and “Insert Copied Cells”.

31	4	Mounting hole position
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Some Generic SCCAFs have completed the Generic SCCAF Assessment available on the [Global FMA Website - SCCAF](#) and have been published in FEDE. These SCCAFs may include:

- An additional column for the “recommended process controls” as even if Generic SCCAF has Process Control Method column completed, the Generic stated method should only be taken as a recommendation. This column must be deleted before approval of the SCCAF.

PFMEA		Process Control Method Reference Information
PFMEA Class	Process Control Method	Delete these columns (AM_AX) before Approval of Program Specific SCCAF

- **Green** text with explanation of what is required is included where Program Application teams need to add or delete information.

No.	Characteristic Description	Specification & Tolerance
1	A/C Hose - material	Program Application team to add Program Specific Material Specification

No.	Characteristic Description	Specification & Tolerance
1	Homologation Requirements	USA - FMVSS 205 China - CCC Europe - ECE43 (Program Application team to delete non-applicable requirements)

- For all Specification & Tolerances that have “xx ± xx” listed then the program application engineer must add the required values and replace the “xx”.

Motor Crank Nut Torque	xx Nm ± xx Nm
Mounting hole diameter	xx mm +xx mm, -xx mm

Critical Characteristics (CC) must be identified on a 2D Drawing or 3D CAD Model if using 3D PMI (no 2D drawing) and the part number identified as an Inverted Delta (CC) part (image 1).

The process for this is detailed in the Cad Drafting Standard [E-2 Special Controls](#) which is part of [FECDS - Ford Motor Company Engineering CAD & Drafting Standards](#) (image 2).

Engineering teams should be communicating the Critical Characteristics identified in the SCCAF and ensuring these are transferred to the drawing / CAD model for all components including those components which are designed by Supplier and where the Supplier completes the 2D drawing (image 3).

All Critical Characteristics identified in the Internal SCCAF should be included on the drawing for that component as per External SCCAFs or for Vehicle Assembly Internal SCCAFs (VME) then they are cascaded to the Illustration for that commodity (image 4).

1

INFORM TO: RESTRICTED SUBSTANCE MANAGEMENT STANDARD P9999-A1 TO SAFEGUARD HEALTH, SAFETY AND THE ENVIRONMENT					
IN ACCORDANCE WITH FORD MOTOR COMPANY ENGINEERING CAD AND DRAFTING STANDARDS VERSION 3.0					
CAD TYPE K-CATIAS	CAD LOC TC	CAD FILE MB3C-1125-D-DWG-01/10	3RD ANGLE PROJ DIMENSIONS ARE IN MILLIMETERS		
PLANT CODE N/A	LINE CODE N/A	OPER NO. N/A	BT NO. N/A	STATION N/A	SIZE N/A
PLANT NAME N/A	DEPT NO. N/A	DESIGN LCHEN136	SCALE 1:1	SHT OF 1	
TITLE/PART NAME DSC BRK WHL					
DRAWING/PART NO. MB3C-1125-DA					
FORD MOTOR COMPANY					

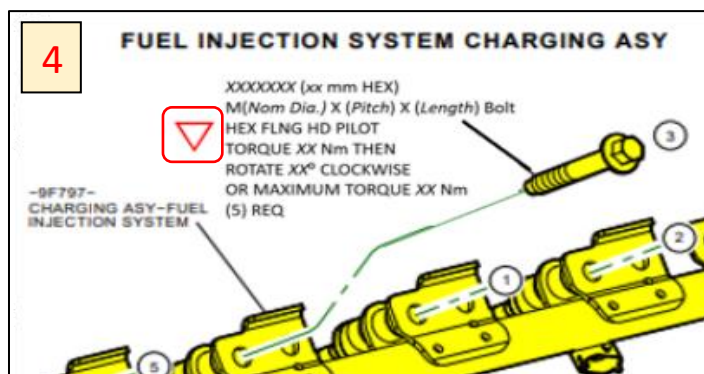
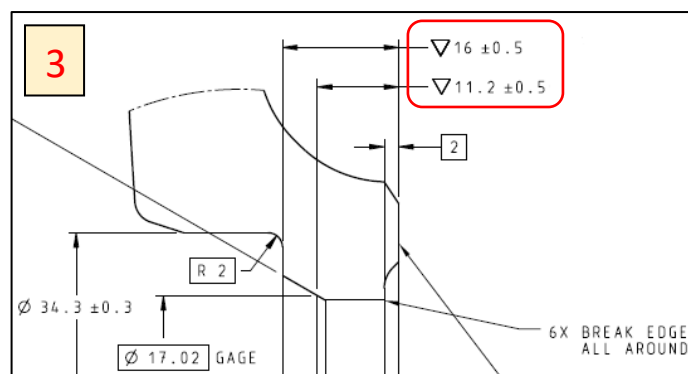
2

Ford Motor Company
Engineering CAD & Drafting Standards

Version: 30
FECDS: E-2

3.2 Inverted Delta Symbol Application

The inverted delta (∇) symbol is applied to features identified as Critical Characteristics on Product Engineering detail and assembly drawings, installation drawings, welding specifications, Engineering Specifications, Engineering Material Specifications, Manufacturing Engineering drawings, process sheets, gauge and fixture drawings. Whenever a Critical Characteristic is identified with the inverted delta (∇), the part, fixture or gauge becomes a Control Item and the part number, fixture number or gauge number is prefixed with the inverted delta (∇) symbol. The inverted delta (∇) is not a portion of the part number, it is a prefix to that number.



Significant Characteristics (SC) should **not** be included on the Ford released drawing / CAD model as per the CAD drafting standard. The actual dimension and tolerance for SCs should appear on a drawing this could be either the Ford drawing or a Supplier Drawing to allow traceability through to Control Plans.

This is for several reasons, these include:

- Once a supplier proves process capability and requests the removal of an SC through the SREA process then a drawing update would be required. This is added cost and workload to the teams.
- If component is made to print in several manufacturing locations or by different suppliers then the number of SCs may vary due to the capability of each manufacturing location / supplier. A good supplier with an in-control processes may only have a small number of SCs but a new supplier or supplier that has quality issues may have many more due to them not being able to show process capability for those specific Special Characteristics.

3.5 Significant Characteristic Identification

The engineering requirements for identifying and communicating Significant Characteristics are defined in Ford Automotive Procedure [FAP03-111](#).

Note: Significant Characteristics are not to be identified on released drawings, details and assemblies. Ford Motor Company drawings must be released with Level One Materials only. Control Plan and DFMEA/PFMEA documents are considered Level Two Materials and as such, information from these documents must not be shown on Level One drawings. For more information regarding Level One and Two Materials, see the PPGTC:

A new help file on the “Life of a Special Characteristic” has been created and loaded onto the website. This has been reviewed and approved by STA and has been uploaded to the STA webpage for use by both the STA team and Suppliers.

[Life of a Special Characteristic - External Link](#) & [SCCAF Webpage](#)

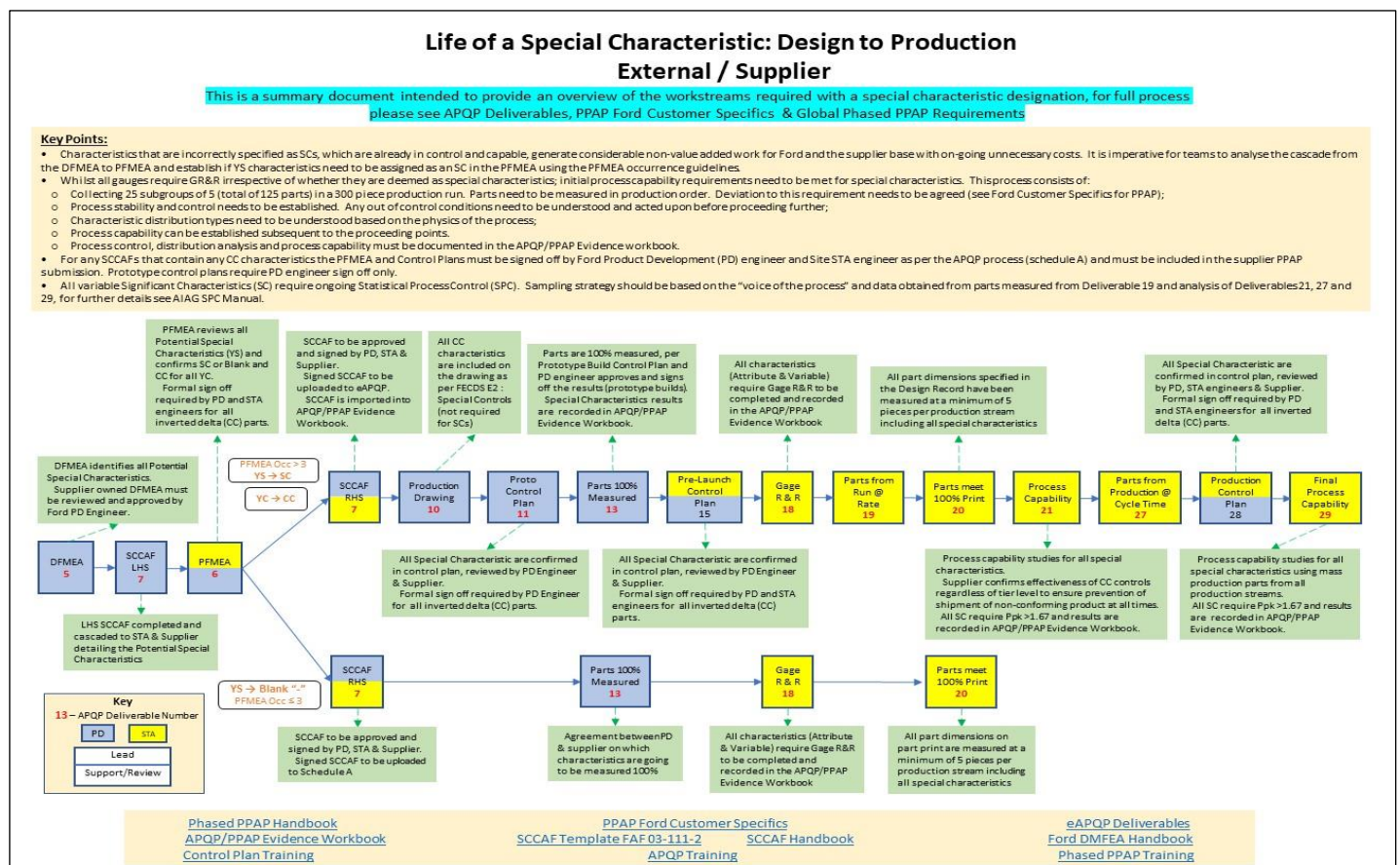
This file details all the steps Special Characteristics go through, where they are managed and controlled to the Production Control Plan.

These steps are from the identification through the DFMEA process to the cascade through the SCCAF and the transfer into the Control Plans and process capability.

This is a summary document intended to provide an overview of the workstreams required with a special characteristic designation, for the full process please see [APQP Deliverables](#), [PPAP Ford Customer Specifics](#) & [Global Phased PPAP Requirements](#).

Key Points:

- Characteristics that are incorrectly specified as SCs, which are already in control and capable, generate considerable non-value-added work for Ford and the supplier base with on-going unnecessary costs. It is imperative for teams to analyse the cascade from the DFMEA to PFMEA and establish if YS characteristics need to be assigned as an SC in the PFMEA using the PFMEA occurrence guidelines.
- Whilst all gauges require GR&R irrespective of whether they are deemed as special characteristics; initial process capability requirements need to be met for special characteristics
- For any SCCAFs that contain any CC characteristics the PFMEA and Control Plans must be signed off by Ford Product Development (PD) engineer and Site STA engineer as per the APQP process (schedule A) and must be included in the supplier PPAP submission. Prototype control plans require PD engineer sign off only.
- All variable Significant Characteristics (SC) require ongoing Statistical Process Control (SPC). Sampling strategy should be based on the “voice of the process” and data obtained from parts measured from Deliverable 19 and analysis of Deliverables 21, 27 and 29, for further details see AIAG SPC Manual.



Once the SCCAF has been fully approved at V2 / Unit TPVDJ the SCCAF will be imported into the “APQP/PPAP Evidence Workbook” which is used to document all measurement data, gauging, capability information related to each characteristic through the launch process.

[APQP/PPAP Evidence Workbook](#)[APQP PPAP Evidence Workbook Statistical Reference Guide](#)[APQP Schedule A Deliverables for GPDS](#)

The APQP / PPAP Evidence Workbook enables common evidence for APQP/PPAP reporting, the use of GPDS for key launch deliverables and improved transparency of results. Leading to faster, better, globally common launch readiness assessments and results.

The Schedule A Deliverable related to SCCAF and the APQP / PPAP Evidence Workbook is Deliverable 7, “Special Characteristics Communication and Agreement Form (SCCAF) Sign off”. Which includes both SCCAF signoff and the start of the APQP / PPAP Evidence Workbook.

Schedule A Deliverable #7 – Special Characteristics Communication and Agreement Form (SCCAF) Sign off

- Special Characteristics have been agreed upon, consistent with the Ford internal procedure FAP03-111 and documented and cascaded to affected suppliers
 - Demonstration of compliance to deliverable #7 for external suppliers is to use the standard SCCAF template (FAF03-111-2) which must be completed/approved and then imported into the APQP/PPAP Evidence Workbook (standardised format/evidence). Both documents must be uploaded into eAPQP against deliverable 7.
 - Deliverable #7 Expectations as per e-APQP (always check e-APQP for latest expectations):
- If the supplier is design responsible (has DFMEA Ownership) then the supplier co-leads the completion of all elements of the Special Characteristics Communication and Agreement Form (FAF 03-111-2) and approves the completed SCCAF in conjunction with the Ford Product Development (PO) engineer, site STA Engineer and Craftsmanship engineer if required. The SCCAF then must be imported into the APQP/PPAP Evidence Workbook. The completed / approved SCCAF and the APQP/PPAP Evidence Workbook must be uploaded into eAPQP and made available to Ford upon request. If Ford is design responsible (has DFMEA Ownership) then the supplier must review and agree the Potential Critical / Significant Characteristics NC and Y SS) identified and Eads the completion of all process related elements of the Special Characteristics Communication and Agreement Form (FAF 03-111-2). The supplier approves the completed SCCAF in conjunction with Ford Product Development (PD) engineer, site STA Engineer and Craftsmanship engineer if required. The SCCAF then must be imported into the APQP/PPAP Evidence Workbook. The completed / approved SCCAF and the APQP/PPAP Evidence Workbook must be uploaded into eAPQP and made available to Ford upon request.
 - Supplier ensures that the Special Characteristics are linked from the available CAD/design/engineering specifications, DFMEA, preliminary Process Failure Mode Effects Analysis (PFMEA) and Prototype/Pre-Launch Control Plans Linkages are recorded in compliance with the Ford Special Characteristics Communication and Agreement Form (FAF-03-111-2).
 - For Powertrain programs, supplier should return the completed Pass Through Characteristics Form PTP07-133f1 with their PTC control methodology to their Component Program Module Team (CPMT) Engineer for approval by the Unit DC milestone, as application.
 - Supplier cascades all applicable Special Characteristics to their supply base and ensures process controls are identified.
 - For appearance parts that include colour, texture and gloss, supplier should identify Special Characteristics per Decorative Component Approval Process (DCAP) Manual and Paint Colour Control Guideline (step#1 and #2). Contact Craftsmanship/Appearance activity for any clarifications.

The Characteristics page of the APQP/PPAP Evidence Workbook may look like a SCCAF but it is NOT a SCCAF. It is designed to import the characteristics from the SCCAF and allow for additional information about the characteristics (e.g. tolerance type, lower & upper spec limits, manufacturing operation identifiers etc.), which flow through to subsequent sheets in the APQP/PPAP Evidence Workbook.

There is functionality that allows additional line items to be entered under the name of “Other” if the teams want to measure additional items that have not been identified as “Special Characteristics”.

Characteristics Details				Data Entry Status				Characteristics Entry in progress							
Complete Characteristic Entry and Lock Characteristics			pplier Code: A1234	Part Number: MB3G-8D000-AA		Build Event: DCV Prototype Build									
			pplier Name: John Doe Inc. - Pleasantville Plant - South Africa												
No.	Source	Characteristic Description	Specification & Tolerance	Type	Lower Spec	Upper Spec	DFME A Class	PFME A Class	Process Control Method	On Control Plan (Y/N)	Manufacturing Operation Identifier	Pass Through Characteristic (PTC) (Y/N)	Error Proofed (Y/N)	Characteristic dependent on sub-supplier (Include PTC) (Y/N)	Notes
1	SCCAF	Port Diameter	67+/-0.3 mm (port A / inlet)				YS	SC	Sub-supplier: Vernier caliper with SPC						
2	SCCAF	Surface Finish	Rz max 16 µm (port A / inlet), Axial				YS	SC	Sub-supplier: Roughness machine with SPC						
3	SCCAF	Flatness for seating	0.1mm Max				YS	SC	CMM with SPC						
4	SCCAF	Motor Crank Nut Torque	25Nm ± 2Nm				YC	CC	100% torque control						
5	SCCAF	Mounting hole position	X 345 Y 150 Z 68 Positional tolerance Ø1.5mm				YS	SC	DC nut runner with Interlock						
6	SCCAF	Bush Push In Force	8Nm to 12Nm				YS	SC	100% in process with load cell						
7	SCCAF	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm				YS	SC	Paint Thickness Ultrasound Tester with SPC						
8	SCCAF	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V				YS	SC	100% EOL voltage tester						
9	Other														

Characteristics Details				Data Entry Status				Characteristics Entry in progress							
Complete Characteristic Entry and Lock Characteristics				Part Number:				Build Event: DCV Prototype Build							
Supplier Code: A1234				Supplier Name: John Doe Inc. - Pleasantville Plant - South Africa											
No.	Source	Characteristic Description	Specification & Tolerance	Type	Lower Spec	Upper Spec	DFME A Class	PFME A Class	Process Control Method	On Control Plan (Y/N)	Manufacturing Operation Identifier	Pass Through Characteristic (PTC) (Y/N)	Error Proofed (Y/N)	Characteristic dependent on sub-supplier (include PTC) (Y/N)	Notes
1	SCCAF	Port Diameter	67+/-0.3 mm (port A / inlet)	Variable	+66.70	+67.30	YS	SC	Sub-supplier: Vernier caliper with SPC	Y	17	Y	N	Y	Characteristic controlled at tier 2
2	SCCAF	Surface Finish	Rz max 16 µm (port A / inlet), Axial	Variable	0	+16.00	YS	SC	Sub-supplier: Roughness machine with SPC	Y	40	Y	N	Y	Characteristic controlled at tier 2
3	SCCAF	Flatness for seating	0.1mm Max	Variable	0	+0.10	YS	SC	CMM with SPC	Y	18	N	N	N	
4	SCCAF	Motor Crank Nut Torque	25Nm ± 2Nm	Variable	+23.00	+27.00	YC	CC	100% torque control DC nut runner with Interlock	Y	36	N	Y	N	
5	SCCAF	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.5mm	Attribute			YS	SC	100 % go / no go fixture	Y	22	N	Y	N	
6	SCCAF	Bush Push In Force	8Nm to 12Nm	Variable	+8.00	+12.00	YS	SC	100% in process with load cell	Y	48	N	Y	N	
7	SCCAF	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	Variable	+20.00	+35.00	YS	SC	Paint Thickness Ultrasound Tester with SPC	Y	52	N	N	N	
8	SCCAF	Alternator voltage	14.4 - 14.6 V with requested voltage of 14.5V	Variable	+14.40	+14.60	YS	SC	100% EOL voltage tester	Y	62	N	Y	N	
9	Other	Shipping Position	Moving glass latched to fixed glass for shipping	Attribute					100% location check with sensor at final assembly fixture	Y	65	N	Y	N	
10	Other														

For more information see the [APQP/PPAP Requirements for GPDS Webpage](#), this webpage includes the training material ([APQP/PPAP Evidence Workbook Training](#)) and other material related to APQP/PPAP requirements.

Examples - External SCCAF Line Items - Completed

[Contents Page](#)

The below table shows examples of Special Characteristics that have been populated on both sides of the External SCCAF.

These examples can be used to understand the minimum standard of information required in the SCCAF document.

They show different scenarios including YS to Blank and the difference in level of control required between an SC and a CC. Each SCCAF must be developed and completed with the support of a cross functional team and-needs to be signed by all the required signatories.

DFMEA				PFMEA		Process Control Method Explanation
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC	CC	Material Certification per batch with material usage controls at each manufacturing station	- This control confirms that the correct incoming material has been supplied to the manufacturing location and each subsequent individual manufacturing station in the event that a site handles more than one type of material. - An SC control can be the same as a CC as it confirms compliance for the complete batch.
2	Material Properties (Flammability - FMVSS 302)	Carpet Surface : WSS-M8P16-B. Foam Underlay : WSS-M99P32-D1 Insulator : WSS-M99P32-D1	YC	CC	Material Certification per batch with material usage controls at each manufacturing station	- Flammability is usually a yearly test and therefore would not be included in the SCCAF. A SCCAF is for ongoing production controls only. - Ongoing controls of Material Properties would be included in the SCCAF and this will ensure compliance to regulations through all of production with confirmation testing at required intervals.
3	Motor Crank Nut Torque	25Nm ± 2Nm	YC	CC	Tier 2: 100% torque control DC nut runner with Interlock Tier 1: Incoming Inspection : 10pcs / batch Torque wrench	Shows Tier 1 is also completing batch sampling, but the control that meets the CC criteria is being completed by the Tier 2.
4	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YC	CC	3D CMM measurement with Batch & Hold (First / Last)	- First off /Last off is acceptable for an agreed stable process where KPIVs need to be monitored. - Stamping examples such as this tend to have very little variation. If the machines / facilities are set up correctly then the components will be stable through the shift. - This control must be used in conjunction with a batch and hold process. Using rational sampling.
5	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	SC	CMM with SPC	- Variable measurements provide continuous data and if sampled appropriately are very powerful in showing shift and drift in processes. - Should not include the frequency or sample size, as this will be established during part development through the vehicle builds.
6	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	SC	100 % Go / No Go fixture	"Discrete" (Go /No Go Gauge) does not give data for SPC therefore Go / No Go should be 100%. (see point 5 - p34).
7	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	-	CMM PPK=3.45	Shows the existing Process Control Method and also the justification of YS to Blank by having a Ppk >1.33 for ongoing production.
8	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	-	Vernier Caliper PFMEA Occurrence = 3	Shows the existing Process Control Method and also the justification of YS to Blank by having the actual occurrence rating from the PFMEA for the specific manufacturing location.
9	Self-Piercing Studs to Mating Parts	Joint quality as per Engineering Spec WI961: Push-Out/Pull-Out Load: 1.0KN Min. Torque-to-Turn: 30 Nm Min	YS	SC	Tier 2: Part sensors controlling insertion depth built into process. Error proofing sensor checks part. Weld teardown Push-Out & Torque (to failure) performed on first and last parts per shift, with Batch and Hold.	- Assembly fixture will ensure parts are loaded and correct insertion depth - Weld teardown by checking "first off" will ensure machine set up is correct and "last off" to make sure no variation through the batch. - The batch and hold means no parts are shipped to Ford before the last part is checked.

Examples - External SCCAF Line Items – Completed (cont.)

[Contents Page](#)

DFMEA				PFMEA		Process Control Method Explanation
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	
10	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	YC	CC	Paint Thickness Ultrasound Tester with Batch & Hold	- Paint process should be stable once correctly set up and KPIVs need to be monitored. - First off / Last Off per shift will enable detection of any drift or major issues with the process. - The measurement should include multiple locations as identified through the set up process and detailed in the Process Control Plan. - Lots need to be quarantined or traceable so that if an issue is found the suspect batch can be contained. Use rational sampling.
11	Surface Coating (Corrosion Protection)	E-Coat/Powder coat as per WSS-M2P177-C2 paint thickness min. 20 µm - max. 35 µm	YS	SC	Paint Thickness Ultrasound Tester with SPC	- As line 10 but rational sampling with SPC will enable detection of any drift or major issues with the process. - Frequencies should not be identified in the SCCAF as this will be established during part development through the vehicle builds. - Frequency and sample size will be dependent on the variation of parts manufactured during the later build phases.
12	Surface Quality / Appearance / Colour	AAR Master Sample / Boundary Book	YS	-	100% Visual "match-to-master" by operator AAR Master Sample PFMEA Occurrence = 3	- Appearance items are: all interior, exterior, luggage compartment, and select under-hood components which are visible to the customer. - AAR sign-off ensures compliance with Ford Colour Harmony Requirements. - Visual "match-to-master" is the specified requirement for AAR sign-off. - These should never been classified as an SC as they are not controllable by physical measurement. - Appropriate process controls should be in place through the Process Control Plan which include controls for (but are not limited to) contamination, moulding machine parameters, dimensional measurements etc. These are the KPIVs that will ensure a part meets its AAR requirements.
13	Edge Radius (Head Impact)	min R5	YC	CC	Poka Yoke by tooling. Tooling sign-off.	- Radius is set in the tooling and will not vary below the required specification so is therefore a Poka-Yoke control. - Poka-Yoke Controls can be implemented as elements of the manufacturing process as well as features incorporated into the product design.
14	Microphone input audio level and distortion	Mic output signal (MIC_OUT) checked against input signal (MIC_IN); EOL PASS/FAIL Parameters / limits are defined in the EOL test software: Tcu_FactoryInstaller_*.*.exe (review latest version)	YC	CC	100% EOL functional test per Ford delivered tester design - PASS/FAIL Test name: Fetch shallow loopback audio .	- EOL test is designed to check for the required signals specified in Spec & Tolerance. 100% check meets the CC Criteria.
15	Welding Specification	WSBCPA-1B310-AA	YC	CC	Full-Automatic Welding. Cut & Etch Test. First Off - Last Off. Cut section for every 50mm with batch & hold per stream/fixture of production.	- Cut & Etch test will ensure welding is to specification and by checking "first off" will ensure machine set up is correct and "last off" to make sure no variation through the batch. - The batch and hold means no parts are shipped to Ford before the last part is checked. - Process Characteristics must be controlled to ensure consistent process outcomes (for more information see p32).
16	Hazardous Material	Control of materials that are hazardous shall follow WSS-M99P9999A	This does not need to be included on the SCCAF. The materials are loaded to the IMDS database at the start of the program and are not an ongoing process control.			

Examples - Internal SCCAF (VOME) Line Items – Completed

[Contents Page](#)

The below table shows examples of Special Characteristics that have been populated on both sides of the Internal Ford Manufactured and Assembled Content SCCAF.

These examples can be used to understand the minimum standard of information required in the SCCAF document. They show different scenarios including YS to Blank and the difference in level of control required between an SC and a CC.

Each SCCAF must be developed and completed with the support of a cross functional team and-needs to be signed by all the required signatories.

For more examples see “[Internal SCCAF Examples](#)” file.

No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	GSPAS / VPS Sheet #
1	Fixing: Horn Bracket Assembly to Body Part Number: W500212-S442 CPSC-Seq: EE00-130601-734 (Full CPSC-Seq required Letters & Numbers) Size: M6 Qty: 1 Process: From: JX6T-13801-** To: JX7B-A16E183-** Remarks: Horn to side member front assembly LHS	Torque: 10.5 Nm +/- 1.6 Nm Remarks: N/A	YC	CC	Machine / Device / Fixture: Electric Tool (DC) Evaluation Measurement Technique: Electrical Tool (DC) & Torque Surveillance as per Global Procedure VOPFAG-241 Control Method: ISPC - Interlocked to Line & QLS Frequency: 100% & Torque Surveillance as per Global Procedure VOPFAG-241	FEG00100
2	Fixing: Battery Box Assembly to Battery Tray support Part Number: W719795-S439 CPSC-Seq: EU00-010605-044 Size: M6 Qty: 3 Process: From: JX6T-10723-** To: JX6T-A16H002-** Remarks: Battery Box to Battery Tray support on front structure	Torque: 10.5 Nm +/- 1.6 Nm Remarks: N/A	YS	SC	Machine / Device / Fixture: Pneumatic Tool Evaluation Measurement Technique: Torque Surveillance as per Global Procedure VOPFAG-241 Control Method: SPC - Torque Control Plan Frequency: Torque Surveillance as per Global Procedure VOPFAG-241	FEB00100
3	Fixing: A-Pillar Grab Handle to A Pillar Part Number: W708669-S303 CPSC-Seq: XXXX-XXXX-XXX (Full CPSC-Seq required Letters & Numbers) Size: M6 Qty: 2 Process: From: 2131406 To: 21312A48 Remarks: Grab Handle To A-Pillar	Torque: 5.4 +/- 0.9 Nm Remarks: N/A	YS	-	PFMEA Occurrence <=3 Machine / Device / Fixture: Pneumatic or Electric Tool	XXXXXXXXXX
4	Connection: Quick Connect (QC) joint assembly Part number: XXX CPSC-Seq: XXX Size: XX Qty: XX Process: From: XXX To: XXX Remark: xxx xxx xxx ***ONE LINE per BASE to BASE connection that are different***	Quick Connect (Fully Seated)	YS	SC	Machine / Device / Fixture: Manual Operation Evaluation Measurement Technique: Manual Operation Control Method: Visual inspection Frequency: 100% Additional: Process GSPAS Sheet: 100% Operator Feel / Audible click Push->Click-> Pull GMR Standard: GMR-1399 QCM Standardised Inspection: GSPAS BOP FNZ 05000	XXXXXXXXXX
5	Connection: Quick Connect (QC) joint assembly - Liquid Fuel Part number: XXX CPSC-Seq: XXX Size: XX Qty: XX Process: From: XXX To: XXX Remark: xxx xxx xxx ***ONE LINE per BASE to BASE connection that are different***	Secondary latch fully engaged (Secondary latch is flush to connector body)	YC	CC	Machine / Device / Fixture: Manual Operation Evaluation Measurement Technique: Manual Operation Control Method: QLS-VO Positive Buy Frequency: 100% Additional: Process GSPAS Sheet: Push-Click-Push-Pull Secondary (Redundant) latch fully engaged	XXXXXXXXXX
6	Equipment: Transmission Fluid Level Process: Oil Filling Part Number: WSS-M2C949-A	1.025L +/-0.025L x.xL +/-y.yL (Top off volume to be confirmed after TT build oil level study, Preliminary Top Off volumes to be provided for TT)	YS	SC	Machine / Device / Fixture: Transmission Fluid Fill Facility Evaluation Measurement Technique: Fill Volume Sample (Beaker) Control Method: SPC - IMR (WBDM) Frequency: Once per shift Additional: GMR Standard: GMR-1454	N/A

Examples - Internal SCCAF (VOME) Line Items – Completed (cont.)

Contents Page

No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	GSPAS / VPS Sheet #
7	Electrical Connection: Horn connector properly seated LHS Harness base part number: 14290 Connector Number: C1RH02-A Qty: 1 Process: From: 14290 To: 130601 Remarks: N/A	Connection: Socket connection fully engaged Remarks: Plastic to Plastic Physical Connection	YS	-	PFMEA Occurrence =< 3 Machine / Device / Fixture: Manual Operation	FWE00600
8	Electrical Signal: ABS Sensor Front (LH/RH) Connection Part number: GN15-2C204-AA Harness base part number: 14290 Connector Number: C1CA03 & C1CA05 Qty: 2 (1xLHS/ 1xRHS) Process: Connect front ABS sensor to front suspension Remarks: Connect front ABS cable connector to engine bay harness	Signal: No DTCs present in the ABS Module Remarks: None of the following DTCs present DTC 0x00037 - Poor Connection DTC 0x0003A - Poor Connection DTC 0x12345 - description (usually open circuit DTCs)	YS	SC	Machine / Device / Fixture: EOL Test verification interlocked with QLS Evaluation Measurement Technique: ABS Self test Control Method: ABS collect Codes Frequency: 100%	N/A
9	Functional Test: Horn function or Horn sound check Part number: JX6T-13801-A* & JX6T-13802-A* Harness base part number: 14290 Connector Number: C1RH02-A & C1RH02-B Qty: 2 (1xLHS/ 1xRHS) Process: N/A Remarks: N/A	Test: Check that horn sound is available when active Remarks: Activate function with below DID and check for DTC 0x12345 or perform current base test with values below If DTC not available the supply Current Base Test characteristics: Nom: 2.5A Min: 2A; Max: 3A DID 0x41D3 - in BCM - Horn Function on/off DID Read characteristics for function on/off including module to be triggered	YS	SC	Machine / Device / Fixture: EOL Test verification interlocked with QLS Evaluation Measurement Technique: CBT for horn function (VOPFAG-268) Control Method: CBT test per VOPFAG-268 Frequency: 100%	N/A
10	Electrical Connection: PIFA Antenna connection Part number: JX7T-19K351-A CPSC-Seq: e.g.150200 (Full CPSC-Seq required Letters & Numbers) Process: e.g. From: 14290 To: 130601 Connector Part Number: FU5T-19C042-A* Connector Number: e.g. C1RH02-A Qty: 1 Remarks: Cellular Antenna detection via OBD2 (diagnostics port)	Signal: No of the following DTCs from EM Remarks: DTC B1246-13 - GSM Antenna, Circuit Open (Socket connection fully engaged and latched)	YS	SC	Machine / Device / Fixture: EOL Test verification interlocked with QLS Evaluation Measurement Technique: EM Self test Control Method: EM collect Codes Frequency: 100%	N/A
11	Electrical Signal: BMS Battery Voltage Part Number: PZ3T-10C652-A*/B* Circuit No: J1-1 ECU ID: 0x726 (BCM) Data Identifier: 0x4028 Qty: 1 Remark: Battery State of Charge (SoC)	12.5 ± 1V >=71% (Note: Example spec provided. PD to define actual spec in engineering phase)	YS	SC	Machine / Device / Fixture: EOL Test System Evaluation Measurement Technique: Battery Monitoring System (BMS) Control Method: QLS ATE Buy OFF Frequency: 100%	N/A
12	Electrical Connection: Passenger Airbag Part Number: PAB (JX7B-A044A74-A(*)) Connector Part Number: e.g. AU5T-19C041-SAB Connector Number: e.g. C1RH02-A CPSC-Seq: e.g.150200 (Full CPSC-Seq required Letters & Numbers) Process: Connector (9U5T-14489-PAA (90 deg.)) to the inflator connector (9U5T-14B364-AA) Connector Number: e.g. CT40053 Remarks: FCI AK2 Code 1 connector, click sound when connected Orientation: 90deg	Signal: None of the following DTCs are set in RCM Remarks: RoW: 0x8010-13 (two wires) FNA: 8010-13, 8011-13 & 91CF-13 (six wires)	YC	CC	Machine / Device / Fixture: EOL Test System Evaluation Measurement Technique: RCM self test & collect code test (DTC read) Control Method: QLS ATE Buyoff Frequency: 100% Additional: Process GSPAS Sheet: Push-Click-Push GMR Standard: HC GMR-1937, 1306	N/A
13	Part Scan: Passenger Airbag Part Number: X7B-A044A74-A(*) Remarks: Correct Part Selected	As Per WERS Release	YC	CC	Machine / Device / Fixture: Barcode Scanner Evaluation Measurement Technique: Scan Control Method: ISPC - Interlocked to Line & QLS Frequency: 100% Additional: Procedure Standard: VOPFAG-250: Error Proofing	N/A
14	Traceability: Passenger Airbag Part Number: X7B-A044A74-A(*) Remarks: Record PAB Serial Information to VIN	Data logged in Trace System Remarks: Serial Number available	YC	CC	Machine / Device / Fixture: Barcode Scanner Evaluation Measurement Technique: Scan to CDCII Control Method: ISPC - Interlocked to Line & QLS or CDCII Check Station Frequency: 100% Additional: GMR Standard: GMR 2133	N/A

Examples - Internal SCCAF (PT) Line Items – Completed

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The below table shows examples of Special Characteristics that have been populated on both sides of the Internal Ford Manufactured and Assembled Content SCCAF.

These examples can be used to understand the minimum standard of information required in the SCCAF document.

Each SCCAF must be developed and completed with the support of a cross functional team and-needs to be signed by all the required signatories.

For more examples see “[Internal SCCAF Examples](#)” file.

No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method	GSPAS / VPS Sheet #
1	Engine Assembly: Tighten Main Bearing Bolts	Torque: 68-115Nm + Angle of 45° Max	YS	SC	Example 1: 100% torque + angle control - DC Nut Runner with trace monitoring, data stored (Q-Worx) Example 2: Machine/Device/Fixture: DC Nut Runner Evaluation Measurement Technique: DC Nut Runner - Trace Monitoring, Data stored in Manufacturing Data Management System (Q-Worx) Control Method: 100% monitored for torque/angle	N/A
2	Cyl head: Leak rate: Water passages (left and right water jacket, and water outlet)	12 CC/MIN AT 138 Kpa	YS	SC	100% Leak Test, Data stored in Manufacturing Data Management System (QLSCM)	NA
3	Crankshaft Heat Treat - Journal Laser Hardening Zone Hardness	45 HRC Min (as specified on Print LC3E-6300-JA sheet 5, view S and T) Poor Example: As per LC3E-6300-JA (needs section info / view on Print)	YS	SC	Lab Check using Rockwell Hardness Tester (variable data) First and last sampling, with a Batch and Hold Process data monitored using manual SPC (charting) Poor Example: Lab check (need to define actual process control method)	NA
4	Crankshaft Balance	18 gm-cm MAX	YS	SC	100% gauged at OP200 CBI Inline Balance Poor Example: Annual Correlation to company master per ES FT4E-6303-BB, section IIIC (Shouldn't include annual checks or routine maintenance on SCCAF)	NA
5	Crankshaft Cleanliness	ES7T4E-6B616-AA, Frame 4, Table 1: For oil ways 1mm max particle size and max amount of foreign matter 6mg; 10mg total	YS	SC	Lab Millipour Test (destructive test) 1/Chamber SOS + AM data monitored using manual SPC (charting)	NA
6	Crankshaft Fillet Condition After Rolling	Poor Example: Smooth, no dents, fretting, galling, etc. (Should not be a YS as not a measurable Spec & Tol - can communicate using Opt. Mfg. Info Transfer Sheet)			PFMEA Occurrence = 2 Opticline laser scanning Visual inspection using Pencil Lead Test,	NA
7	Trans Case/AT Housing: MLS Mill slot Location [T202]	M66.84 / 68.64mm	YC	CC	First and last sampling, with a Batch and Hold Process RF Tool Control, variable hand gauge, CMM check and birth history traceability Poor Example: variable hand gauge with SPC controls, capable per PPAP process (capability not an acceptable control method for a CC, PPAP is not a control method, SPC is not an acceptable control on a CC)	NA
8	Cyl head machined - valve seat pocket press to full depth	Press to depth (0mm gap)	YS	-	Occurrence = 3 100% Press Force vs Distance Monitoring (Load Cell), Data stored in Manufacturing Data Management System Sight Pipe Gauge 1 part/machine/day	NA
9	Transmission Assembly: Installation - OML Shaft Roll Pin (9L8P-7G100-AA)	Press depth per ED8P-7000-A Section 1.4 (7.8 +/- 1.0)	YC	CC	100% Sony probe verification OL10 Case Sub Assembly OL10.0120.1B 100% checked in Subassy w/ Servo Press OL10 Case Sub Assembly OL10.0120.1B	NA

Special Characteristic

- Product or process characteristics (CC, SC, OS, and HI) that affect vehicle or process safety, compliance with government regulations, customer satisfaction, or process operation. Special Characteristics require Quality Planning actions that must be addressed in a Control Plan.

SCCAF **Special Characteristics Communication & Agreement Form.**

- Form used in the process for Ford Automotive Procedure FAP-03-111 - Selection, Identification, and Control of Special Characteristics.

FMA **Failure Mode Avoidance.**

- A set of analytical tools to do engineering in a disciplined manner and document it for use in the future.

LVEA **Lead Vehicle Engineering Activity.**

- LVEA is the region responsible for overall delivery of the vehicle or platform.
- May be the same or different from the LEA.

LEA **Lead Engineering Activity.**

- LEA is the engineering activity (functional) that has lead for the design of a particular system or component.
- May be the same or different from the LVEA.
- LEA is responsible to ensure the process is completed by the specific Suppliers for the relevant system or component.

STA **Supplier Technical Assistance.**

- Lead in the selection and development of Ford's global supply base to achieve the highest manufacturing capability in support of world class customer satisfaction and safety.

SREA **Supplier Request for Engineering Approval.**

- Ford's process for handling and approving post-Job #1 process changes initiated by Suppliers.

FMEA **Failure Mode & Effects Analysis.**

- Recognize and evaluate the potential failure of a product or process and its effects.
- Identify actions which could eliminate or reduce the chance of the potential failure occurring.
- Document the process.

DFMEA **Design Failure Mode & Effects Analysis.**

- A FMEA that is focused on the design of a product.

PFMEA **Process Failure Mode & Effects Analysis.**

- A FMEA that is focused on the manufacturing/assembly process.

YC / CC **Critical Characteristic (YC – Potential Critical Characteristic).**

- Are product or process characteristics that can affect compliance with government regulations and/or safe vehicle and/or product function.
- Require specific manufacturing, assembly, shipping and/or monitoring action to prevent shipment of non-conforming product and must be included in the Control Plan.

YS / SC **Significant Characteristics (YS – Potential Significant Characteristic).**

- Those product, process, and test requirements that are important for customer satisfaction. Significant Characteristics require Quality Planning actions that must be addressed in a Control Plan.

OS **Operator Safety Characteristics.**

- Operator safety characteristics (OS): Operator Safety Characteristics are related to process parameters or product characteristics that may adversely affect the safety of the operator or compliance with governmental regulations.

HI **High Impact Characteristic.**

- High Impact Characteristics are related to process parameters or product characteristics that can adversely affect the operation of the process or subsequent operations, but that do not adversely impact customer satisfaction.

PCP **Process Control Plan.**

- Control Plans are written descriptions of the systems used to control and minimize product and process variation.
- They specify the process monitoring and Process Control Methods (i.e. Special Controls) used to control Special Characteristics.

KPIV **Key Process Input Variable.**

- A characteristic of a process that determines the performance level of a higher-level key process output variable.

NTEI **New Tool End Item.**

- The parts on a new model year product that have changed since the previous model year.

Ppk **Process Capability (Ppk).**

- Is a measure of how well the process output meets the customer requirements. It takes into account both the spread & centering of the data. Assumptions include rational sampling plan, stability, control & the appropriate data distribution.

GSDB **Global Supplier Database.**

- Companies that conduct business with Ford have a 4 or 5-character GSDB Parent Code this is used to identify the company for purchasing and other Supplier related activities.

GD & T **Geometrical Dimensioning and Tolerancing - [Ford GD & T Information](#).**

- Is a system for defining and communicating engineering tolerances. It uses a symbolic language on engineering drawings and computer-generated three-dimensional solid models that explicitly describe nominal geometry and its allowable variation. .

JV **Ford Joint Venture.**

- Ford partnership such as Changan Ford (CAF) which is a joint venture for vehicle production based in Chongqing, China.

Batch & Hold

- Is a process control method that requires a batch of parts to be put on hold until appropriate sampling can be completed to confirm part quality and prevent the shipment of non-conforming product. There are typically two forms of batch and hold processes (see [p30](#) for more details).

Below is an example SCCAF template with some included errors.

Review the SCCAF and identify the errors included in the sample SCCAF template.

The answers with a description of why and a correct suggestion are contained on the next pages.

For this exercise consider that all specifications listed for that relevant Characteristics are correct, do not consider these as potential errors.

There are 12 errors:

- 2 Header
- 1 Complete line item (consider as one error)
- 3 LHS
- 6 RHS

Graphic showing location of answers is on p50 and detailed description of error and suggested correct completion is shown on p51.

Special Characteristics Communication and Agreement - External					OPTIONAL:	SCCAF Draft Vers:	Date:	
DFMEA & Potential Special Characteristics Type ONLY in Yellow Boxes PFMEA & Special Characteristics Agreement Type ONLY in Green Boxes		SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section SCCAF Handbook - SCCAF Global FMA Website		(Suppliers Only - Accessible through Ford Supplier Portal). (Ford Internal Only)			Special Characteristics for:	
Lead / Support roles and timing for each stage		Ford PD engineer / Supplier		Supplier / Ford PD		Z987 MCA Alan Smith Inc. - Detroit Plant - USA AB3D-9876-A		
Model Year & Program Code: Z987 MCA		Part / System Number: AB3D-9876-A		Ford D & R Engineer Approval / Date:		Approval Signature Name e-mail Address Approval Date		
Supplier Name / Plant: Alan Smith Inc. - Detroit Plant - USA		Engineering Release Number:		Ford STA Engineer Approval / Date:				
Supplier GSDB Code: A1234		Design Lead: Supplier		Ford Craftsmanship Approval / Date:				
Part / System Name: Bracket - Fuel Mount		Ford Customer Plant: Flat Rock - Michigan - USA		Supplier Plant Quality Manager Approval / Date:				
CPSC: 100103		DFMEA Reference: Generic - Fuel Distribution - 100103		Note: Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).				
Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA - Global FMA Website (Ford Internal Only)		From DFMEA: YC Sev 9,10 : Potential CC. YS Sev 5-8 and may require special control to maintain process capability : Potential SC.		From PFMEA: CC Sev 9,10 (Part) : Critical Characteristic. SC Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.		OS Sev 9,10 (Process) : Operator Safety. HI Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.		
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS		YC 4 YS 7		CC 3 SC 6 - 1		OS 0 HI 1		
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>. Special Characteristics Communication and Agreement for all UP content is required by UPV2. Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.								
DFMEA				PFMEA				
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method			
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC	CC	Material Certification per batch with material usage controls at manufacturing stations			
2	Bracket Mount Nut Torque	25 ± 2	YC	CC	100% torque control DC nut runner with Interlock			
3	Fuel Filter Hole diameter	2.5mm +0.0mm, -0.03mm	YS	-	CMM			
4	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	SC	100 % CMM PV only 1 per month			
5	Retaining Ring Push In Force	Design Rule DR-010807-711906	YS	HI	100% in process with load cell			
6	Bracket Surface Coating - Paint Thickness (Corrosion Protection)	min. 20 µm - max. 35 µm	YS	SC	Per Batch for PPAP Annual Review			
7	Compressor voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS	SC	100% EOL voltage tester			
8	Durability Test	RQT-001100-002458	YS	SC	Rig test			
9	Material Properties (Flammability - FMVSS 302)	WSS-M8P16-B	YC	SC	Material Certification per batch with material usage controls at manufacturing stations			
10	Hose Diameter	Ø 12.7mm	YS	SC	Vernier Caliper with SPC			
11	Welding Specification	WSBCPA-1B310-AA	YC	CC	Full-Automatic Welding. Weld Integrity Test - Sectioning - First off / Last Off			

Test Your Knowledge – Answers

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Graphic showing errors on SCCAF template, see next page for detailed description of error and suggested completion information for each error.

FAF03-111-2 v2.1 July 2020				Special Characteristics Communication and Agreement - External				OPTIONAL: SCCAF Draft Vers: Date:			
DFMEA & Potential Special Characteristics		Type ONLY in Yellow Boxes	PFMEA & Special Characteristics Agreement	Type ONLY in Green Boxes	SCCAF Handbook - Global Phased PPAP - Web Site See SCCAF Section SCCAF Handbook SCCAF Global FMA Website		(Suppliers Only - Accessible through Ford Supplier Portal). Special Characteristics for:				
Lead / Support roles and timing for each stage		Ford PD engineer / Supplier		Supplier / Ford PD		(Ford Internal Only)		Z987 MCA Alan Smith Inc. - Detroit Plant - USA AB3D-9876-A			
Model Year & Program Code: Z987 MCA		1		Part / System Number: AB3D-9876-A		Ford D & R Engineer Approval / Date:		Approval Signature Name e-mail Address Approval Date			
Supplier Name / Plant: Alan Smith Inc. - Detroit Plant - USA				Engineering Release Number:		Ford STA Engineer Approval / Date:					
Supplier GSD Code: A1234				Design Lead: Supplier		Ford Craftmanship Approval / Date:					
Part / System Name: Bracket - Fuel Mount				Ford Customer Plant: Flat Rock - Michigan - USA		Supplier Plant Quality Manager Approval / Date:					
CPSC: 100103				DFMEA Reference: Generic - Fuel Distribution - 100103		2		Approvals may be obtained by a physical signature or electronic sign-off. Recommended: Use DocuSign (Some Country Restrictions for External Approval) to obtain signatures as it will provide electronic evidence to store with the completed SCCAF & SCCAF Assessment. (See SCCAF Webpage for DocuSign help & DocuSign Country Restrictions).			
Key: For Special Characteristic definitions refer to Ford FMEA Handbook. DFMEA: Global FMA Website (Ford Internal Only)		From DFMEA: YC Sev 9,10 : Potential CC. YS Sev 5-8 and may require special control to maintain process capability : Potential SC.		From PFMEA: CC Sev 9,10 (Part) : Critical Characteristic. SC Sev 5-8 & Occ 4-10 (Part) : Significant Characteristic.		OS Sev 9,10 (Process) : Operator Safety. HI Sev 5-8 & Occ 4-10 (Process) : High Impact Characteristic.					
Special Characteristic Totals: Note 1: Automatic calculation of totals (for guidance only). Note 2: Count of SC may be less than count of YS		YC 4 YS 7		CC 3 SC 5 - 1		OS 0 HI 1					
Special Characteristics Communication and Agreement for all UN content required to support UN design and development is required by UNV2. The remaining UN content not required to support UN design and development is required by <PA>. Special Characteristics Communication and Agreement for all UP content is required by UPV2. Where Specification & Tolerance differ in SCCAF to latest released data, released data is Master. Note: Process Control Plan should be crosschecked to released data at creation and at each new data release.											
DFMEA						PFMEA					
No.	Characteristic Description	Specification & Tolerance	DFMEA Class	PFMEA Class	Process Control Method						
1	Sheet Steel Material Properties	WSS-M1A365-A21	YC	CC	Material Certification per batch with material usage controls at manufacturing stations						
2	Bracket Mount Nut Torque	25 ± 2 4	YC	CC	100% torque control DC nut runner with Interlock						
3	Fuel Filter Hole diameter	2.5mm +0.0mm, -0.03mm	YS	-	CMM 9						
4	Mounting hole position	X:345 Y:150 Z:68 Positional tolerance Ø1.6mm	YS	SC	100 % CMM PV only 1 per month 10						
5	Retaining Ring Push In Force	Design Rule DR-010807-711906 5	YS	HI 7	100% in process with load cell						
6	Bracket Surface Coating - Paint Thickness (Corrosion Protection)	min. 20 µm - max. 35 µm	YS	SC	Per Batch for PPAP Annual Review 11						
7	Compressor voltage	14.4 - 14.6 V with requested voltage of 14.5V	YS	SC	100% EOL voltage tester						
8	Durability Test 3	RQT-001100-002458 3	YS	SC	Rig test 3						
9	Material Properties (Flammability - FMVSS 302)	WSS-M8P16-B	YC	SC 8	Material Certification per batch with material usage controls at manufacturing stations						
10	Hose Diameter	Ø 12.7mm 6	YS	SC	Vernier Caliper with SPC						
11	Welding Specification	WSBCPA-1B310-AA	YC	CC	Full-Automatic Welding. Weld Integrity Test - Sectioning - First off / Last Off 12						

Error	Description	Completion Suggestion
1 Header	Model Year is missing, need to enter the Model Year and Program Code of the Vehicle or Powertrain program for this part (p17).	2021 B123 MCA
2 Header	Ford DFMEA listed when "Design Lead" is selected as "Supplier." Need to enter unique Supplier DFMEA identification (p17).	John Doe Engineering - DFMEA 1234-9876.
3 Complete Line Item	Whole line item (p19): Durability test is not a valid characteristic as it is not controllable on an ongoing basis. The actual Product Characteristics that results in compliance with the requirement should be defined to determine what must be controlled.	Tensile Strength
4 LHS	Missing Units, all dimensions require the specific unit of measurement (p23).	25 ± 2 Nm
5 LHS	A design Rule is not a valid Specification & Tolerance, it is not a WERS released Ford Document. Should include the actual measurable values identified within the Design Rule (p23).	8 - 12 KN
6 LHS	Missing Tolerance all dimensions require a tolerance (p23).	Ø 12.7mm ± 0.2mm
7 RHS	YS to HI. A YS can only become a "SC" or "Blank –". OS and HI Characteristics relate to the process only, they exist only in the PFMEA – there is no potential OS or potential HI in the DFMEA (p28 & p34).	"YS" → "SC" or "YS" → "-" dependent on PFMEA Occurrence
8 RHS	YC to SC. A YC can only ever become a CC (p28).	"YC" → "CC"
9 RHS	As "YS" → "-" need either the actual Ppk value for that process in that specific manufacturing location or from the PFMEA for the relevant "potential cause" for that specific manufacturing location (p29).	CMM PFMEA Occurrence = 2
10 RHS	PV should not be included on a SCCAF as it should only be the planned Production Process Control Method. Reference to PPAP, DV/PV etc. are not acceptable, even if in addition to an adequate process control. Need to include ongoing Process Control Method not a frequency (p34 & p35).	CMM with SPC
11 RHS	PPAP should not be included SCCAF is to detail planned production process control method. Annual review is not an ongoing control and acceptable for a SC characteristic (p34 & p35).	Paint Thickness Ultrasound Tester with SPC
12 RHS	CCs require Special Controls to prevent shipment of non-conforming product at all times, the control will require a "Batch & Hold" process to ensure non-conforming parts are not shipped between the first and last piece check (p31).	Full-Automatic Welding. Weld Integrity Test - Sectioning - First off / Last Off with Batch & Hold